THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF ENVIRONMENTAL AND EARTH SCIENCES (RSEES)

DEPARTMENT OF GEOLOGY

COURSE STRUCTURE & SYLLABUS

(BASED ON NATIONAL EDUCATION POLICY 2020)

FOR

B.Sc. IN GEOLOGY

(4 YEARS SINGLE MAJOR)

ACADEMIC YEAR

2026-27

B. Sc. Geology

Programme Structure

1st SEMESTER				
Sl. No	**Course Code**	**Name of Courses**	**Course Level**	**Credits**
		Major Courses		
1	GEOL162M101	Physical Geology	100	3
3	GEOL162M102	Mineral Science	100	3
4	GEOL162M113	Mineral Science Practical	100	1
		Minor Course		
5	GEOL162N101	Physical Geology	100	3
		Interdisciplinary Courses		
6	IKS992K101	Introduction to Indian Knowledge System I	-	3
		Ability Enhancement Courses		
7	CEN982A101	Introduction to Effective Communication	-	2
		Skill Enhancement Courses		
8	GEOL162S111	Geological Mapping and Surveying	-	3
		Value Added Courses		
09	GEOL162V101	Earth and Climate Systems	-	3
10		**Course through SWAYAM portal *(to be selected by the department)***	-	3 or 4
			TOTAL	21 + (3 or 4)
2nd SEMESTER				
Sl. No	**Course Code**	**Name of Courses**	**Course Level**	**Credits**
		Major Courses		
1	GEOL162M201	Igneous Petrology and Geochemistry	100	4
3	GEOL162M212	Igneous Petrology and Geochemistry Practical	100	2
		Minor Course		
4	GEOL162N201	Geology and Natural Hazards	100	3
		Interdisciplinary Courses		
5	IKS992K201	Introduction to Indian Knowledge System II	-	3
		Ability Enhancement Courses		
6	CEN982A201	Approaches to Verbal and Non-verbal Communication	-	2
		Skill Enhancement Courses		
7	GEOL162S211	Geomorphology and Landform Analysis	-	3
		Value Added Courses		
8	GEOL162V201	Disaster Management in the Indian Context	-	3
9		**Course through SWAYAM portal *(to be selected by the department)***	-	3 or 4
			TOTAL	20 + (3 or 4)
3rd SEMESTER				
Sl. No	**Course Code**	**Name of Courses**	**Course Level**	**Credits**

		Major Courses		
1	GEOL162M301	Metamorphic Petrology	200	3
2	GEOL162M312	Metamorphic Petrology Practical	200	1
3	GEOL162M303	Sedimentology	200	3
4	GEOL162M314	Sedimentology Practical - I	200	1
		Minor Course		
5	GEOL162N301	Mineralogy	200	4
		Interdisciplinary Courses		
6	GEOL162I301	Physics and Chemistry of Earth	-	3
		Ability Enhancement Courses		
7	CEN982A301	Fundamentals of Business Communication	-	1
8	BHS92A302	Behavioural Science - III	-	1
		Skill Enhancement Courses		
9	GEOL162S311	Remote Sensing and GIS	-	3
10		**Course through SWAYAM portal *(to be selected by the department)***	-	3 or 4
			TOTAL	20 + (3 or 4)
4th SEMESTER				
Sl. No	**Course Code**	**Name of Courses**	**Course Level**	**Credits**
		Major Courses		
1	GEOL162M401	Principles of Stratigraphy	200	3
2	GEOL162M402	Palaeontology	200	3
3	GEOL162M403	Earth Science in Ancient India	200	4
4	GEOL162M414	Stratigraphy and Palaeontology Practical	200	2
		Minor Course		
5	GEOL162N401	Petrology	200	3
6	GEOL162N402	Structural Geology	200	3
		Ability Enhancement Courses		
7	CEN982A401	Business Communication: Concepts and Skills	-	2
8		**Course through SWAYAM portal *(to be selected by the department)***	-	3 or 4
			TOTAL	20 + (3 or 4)
5th SEMESTER				
Sl. No	**Course Code**	**Name of Courses**	**Course Level**	**Credits**
		Major Courses		
1	GEOL162M501	Structural Geology	300	3
2	GEOL162M512	Structural Geology Practical - I	300	1
3	GEOL162M503	Indian Stratigraphy	300	4
4	GEOL162M504	Hydrogeology	300	3
5	GEOL162M515	Hydrogeology Practical	300	1
6	GEOL162M526	Summer Internship	-	4
		Minor Course		
7	GEOL162N501	Fuel Geology	200	4
			TOTAL	20
6th SEMESTER				
Sl. No	**Course Code**	**Name of Courses**	**Course Level**	**Credits**
		Major Courses		
1	GEOL162M601	Geostatistics	300	4
2	GEOL162M602	Economic Geology	300	3
3	GEOL162M613	Economic Geology Practical	300	1

4	GEOL162M604	Engineering and Environmental Geology	300	3
5	GEOL162M615	Engineering Geology Practical	300	1
6	GEOL162M606	Fuel Geology	300	3
7	GEOL162M617	Fuel Geology Practical	300	1
		Minor Course		
8	GEOL162N601	Environmental Geology	200	4
			TOTAL	20
7th SEMESTER				
Sl. No	**Course Code**	**Name of Courses**	**Course Level**	**Credits**
		Major Courses		
1	GEOL162M701	Advanced Structural Geology and Tectonics	400	3
2	GEOL162M712	Structural Geology Practical - II	400	1
3	GEOL162M703	Advanced Igneous and Metamorphic Petrology	400	3
4	GEOL162M714	Igneous and Metamorphic Practical	400	1
5	GEOL162M705	Advanced Sedimentology and Quaternary Geology	400	3
6	GEOL162M716	Sedimentology Practical - II	400	1
7	GEOL162M728	Dissertation research	400	4
		Course in lieu of dissertation		
8	GEOL162M707	Climatology and Oceanography	400	4
		Minor Course		
9	GEOL162N701	Stratigraphy	300	4
			TOTAL	20
8th SEMESTER				
Sl. No	**Course Code**	**Name of Courses**	**Course Level**	**Credits**
		Major Courses		
1	GEOL162M801	Geomorphology	400	3
2	GEOL162M812	Geomorphology Practical	400	1
3	GEOL162M803	Research Methodology	400	4
4	GEOL162M824	Dissertation	-	12
		Advanced Course in lieu of Dissertation		
5	GEOL162M804	Geology of North-East India	400	4
6	GEOL162M805	Planetary Geology	400	4
7	GEOL162M806	Urban Geology	400	4
			TOTAL	20

Detailed Syllabus Of Semester 1

Type of Course: Major	Physical Geology Course Level: 100 Credit: 3 L-T-P-C: 2-0-0-2 Scheme of Evaluation: Theory	Course Code: GEOL162M101

Course Objectives: Physical Geology is an introductory course that covers the fundamental principles of geology. It provides an understanding of the Earth's internal and external processes that shape the Earth's surface and subsurface features. This course covers a range of topics, including mineralogy, petrology, plate tectonics, structural geology, and geological time.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Remember the fundamental concepts of geology, including the rock cycle, plate tectonics, and geological time.	**BT 1**
CO 2	Explain the processes that form the Earth's surface features and analyze the geological structures and their influence on the formation of natural resources.	**BT 2**
CO 3	Apply the principles of mineralogy and petrology to identify and classify different types of rocks and minerals, interpret geological maps and cross-sections, and solve geological problems.	**BT 3**
CO 4	Analyse geomorphological processes and data to understand the various geomorphic activities and their impact on landscape evolution.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	Earth Science & its branches. Origin and evolution of the Universe and Solar System, The standard model of planetary formation. General features of the components of the solar system. Distribution of elements in solar system and in Earth.	10
Unit 2	Mechanical layering of the Earth: lithosphere, asthenosphere, mantle and core. Formation of core, mantle, crust, hydrosphere, and atmosphere. Introduction to Rocks – its types and associated features, Rock cycle, Introduction to the concept of Geological Time Scale.	10
Unit 3	Concept of continental drift, seafloor spreading and plate tectonics. Plate boundaries and their geological effects: origin of oceans, continents, mountains and rift valleys. Earthquake and earthquake belts. Volcanoes and its types, distribution and eruptions. Geothermal gradient and internal heat of the Earth. Earth's magnetic field; Convection in Earth's core and production of its magnetic field.	12
Unit 4	Geomorphological processes and their significance. concept of base level and datum. Concept of exogenic and endogenic processes. Weathering, erosion, mass-wasting and their types. Landforms produced by – glacial processes, fluvial processes, aeolian processes, coastal processes, igneous activities.	12
Experiential Learning: Home Assignments – 8 hrs, Presentation – 4 hrs, Video Screening – 4 hrs		16
Total Credit Hours		**60**

Text Books:

1) Introduction to Physical Geology – Thompson & Turk
2) Essentials of Geology - Stephen Marshak, 4th edition, W. W. Norton & Company

Reference Books:

1) Physical Geology – R. F. Flint and J Skinner, John Wiley and Sons, Inc.
2) Global Tectonics - Philip Kearey, Keith A. Klepeis, Frederick J. Vine, (3rd edition, 2009), Wiley-Blackwell.

Type of Course: Major	Mineral Science Course Level: 100 Credit: 3 L-T-P-C: 2-1-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162M102

Course Objectives: Mineral Science is an undergraduate-level course that focuses on the study of minerals and their properties, including crystallography, optical properties, chemical composition, and physical characteristics. The course provides an overview of the formation and classification of minerals, as well as the processes involved in their identification and analysis.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Remember and identify the properties of minerals, their classification, and crystallographic systems.	**BT 1**
CO 2	Explain the physical and chemical properties of minerals and their economic importance.	**BT 2**
CO 3	Apply mineralogical knowledge to identify minerals using optical microscopy and physical characteristics of the specimens.	**BT 3**
CO 4	Analyse various crystallographic and mineralogical data to identify and classify them into various classes.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	Concept of crystal, crystalline and amorphous substances. Unit cell and Lattice structures, Bravais Lattices; Types of crystal structures (e.g., cubic, hexagonal); Symmetry elements and point groups; Crystallographic axes and planes; Overview of crystal systems (e.g., isometric, tetragonal, etc.); Crystallographic axes and symmetry elements for each system.	11
Unit 2	Interfacial angle, crystal parameters and indices. Stereograms and Hermann- Mauguin System. Relationship between crystallography and mineral properties. Isomorphism, polymorphism, solid solution and exsolution.	11
Unit 3	Minerals - definition, physical and chemical properties; Chemical classification of minerals. Silicate and non-silicate structures of minerals. Study of physical properties of minerals of the following group of minerals: Olivine, Pyroxene, Amphibole, Mica, Silica and Feldspar.	11
Unit 4	Polarization of light, Polarisers. Functions of petrological microscope. Optical behaviour of minerals: Absorption, Transmission and Double-refraction of light. Theory of light propagation in minerals: Isotropy and Anisotropy; Optic axis. Optical properties of minerals in thin section. Introduction to X-Ray diffractometry in minerals.	11
Experiential Learning: Home Assignments – 8 hrs, Presentation – 4 hrs, Video Screening – 4 hrs		16
Total Credit Hours		**60**

Text Books Suggested:

1) Mineralogy - Dexter Perkins, 3rd edition (2015), Pearson Publication.

Reference books:

1) Introduction to Optical Mineralogy – William D. Nesse, 3rd edition (2004), Oxford University Press.
2) The Manual of Mineral Science (after James D. Dana) - Dutrow, B., Dwight, J., & Klein, C.; (2007) J. Wiley & Sons.
3) Introduction to Rock Forming Minerals – W. A. Deer, R. A. Howie, and J. Zussman, 3rd edition (2013), Prentice Hall.

Type of Course: Major	Mineral Science Practical Course Level: 100 Credit: 1 L-T-P-C: 0-0-2-1 Scheme of Evaluation: Practical	Course Code: GEOL162M113

Course Objectives: Mineral Science is an undergraduate-level course that focuses on the study of minerals and their properties, including crystallography, optical properties, chemical composition, and physical characteristics. The course provides an overview of the formation and classification of minerals, as well as the processes involved in their identification and analysis.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Remember and identify the properties of minerals, their classification, and crystallographic systems.	**BT 1**
CO 2	Explain the physical and chemical properties of minerals and their economic importance.	**BT 2**
CO 3	Apply mineralogical knowledge to identify minerals using optical microscopy and physical characteristics of the specimens.	**BT 3**
CO 4	Analyse various crystallographic and mineralogical data to identify and classify them into various classes.	**BT 4**

Modules	Topics and Course Content	Hours
I	Identification of crystal systems and symmetry elements using physical models. Drawing crystal forms; plotting axes and symmetry elements on diagrams. Determination of Miller indices and practice exercises using crystal faces. Plotting of symmetry elements using stereonets. Observation of Physical Properties of Minerals (lustre, cleavage, fracture, colour, streak, hardness, etc.) in Hand-Specimens – Olivine, Garnet, Sillimanite, Kyanite, Staurolite, Tourmaline, Enstatite, Diopside, Augite, Actinolite, Hypersthene, Hornblende, Serpentine, Muscovite, Biotite, Quartz and its varieties, Orthoclase, Plagioclase, Microcline, Nepheline, Sodalite, Calcite, Beryl, Talc, Zeolite. Identification of the parts of Petrological Microscope and demonstration of functioning (polarisers, stage, condenser, etc.). Microscopic examination of common anisotropic minerals - Olivine, Garnet, Sillimanite, Kyanite, Enstatite, Diopside, Augite, Hypersthene, Hornblende, Muscovite, Biotite, Quartz and its varieties, Orthoclase, Plagioclase, Microcline, Calcite. Determination of Optic Axis Determination of Pleochroic Scheme of minerals. Identification of Plagioclase Feldspars by Michel-Levy method.	30
Total Credit Hours		**30**

Text Books Suggested:

1) Mineralogy - Dexter Perkins, 3rd edition (2015), Pearson Publication.

Reference books:

1) Introduction to Optical Mineralogy – William D. Nesse, 3rd edition (2004), Oxford University Press.
2) The Manual of Mineral Science (after James D. Dana) - Dutrow, B., Dwight, J., & Klein, C.; (2007) J. Wiley & Sons.
3) Introduction to Rock Forming Minerals – W. A. Deer, R. A. Howie, and J. Zussman, 3rd edition (2013), Prentice Hall.

Type of Course: Minor	Physical Geology Course Level: 100 Credit: 3 L-T-P-C: 3-0-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162N101

Course Objectives: This course provides an overview of the fundamental principles of physical geology, focusing on the study of Earth's processes, materials, and the dynamic forces that shape the planet.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Demonstrate a comprehensive understanding of Earth's structure, composition, and geological processes.	**BT 1**
CO 2	Comprehend the principles of plate tectonics and the processes that drive geological changes.	**BT 2**
CO 3	Apply geological concepts to explain natural phenomena, such as earthquakes, volcanoes, and mountain formation.	**BT 3**
CO 4	Analyze geological data to interpret Earth's history and the formation of various geological formations.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	Definition and scope of physical geology Earth's structure, composition, and geologic time Study of minerals and their properties Classification and identification of rocks	17
Unit 2	Understanding plate movements and boundaries The role of plate tectonics in earthquakes, volcanoes, and mountain building Understanding seismicity and volcanic eruptions Impact of earthquakes and volcanoes on the Earth's surface	16
Unit 3	Study of weathering, erosion, mass-wasting and sedimentation Fluvial, glacial, marine and coastal processes and features. Epirogenic processes and movements.	17
Unit 4	Geological processes and their influence on the environment Human interaction with geological hazards such as landslides, earthquakes, volcanic eruptions, floods, mining activities and other civil engineering projects. Relationship between geology and climate change.	16
Experiential Learning: Home Assignments – 10 hrs, Presentation – 10 hrs, Video Screening – 4 hrs		24
Total Credit Hours		**90**

Text Books:

1) Introduction to Physical Geology – Thompson & Turk
2) Essentials of Geology - Stephen Marshak, 4th edition, W. W. Norton & Company

Reference Books:

1) Physical Geology – R. F. Flint and J Skinner, John Wiley and Sons, Inc.
2) Global Tectonics - Philip Kearey, Keith A. Klepeis, Frederick J. Vine, (3rd edition, 2009), Wiley-Blackwell.

Type of Course: IDC	Introduction to Indian Knowledge System - I Credit: 3 L-T-P-C: 2-1-0-3 Scheme of Evaluation: Theory	Course Code: IKS992K101

Course Objectives: This Foundation course is designed to present an overall introduction to all the streams of IKS relevant to the UG program. It would enable students to explore the most fundamental ideas that have shaped Indian Knowledge Traditions over the centuries.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall the rich heritage of Indian knowledge systems	**BT 1**
CO 2	Describe the contribution of Indian knowledge systems to the world	**BT 2**
CO 3	Demonstrate knowledge of sociocultural and ethnolinguistic diversity that constitutes the soul of Bharatvarsha	**BT 2**
CO 4	Apply traditional knowledge and techniques in day-to-day life	**BT 3**
CO 5	Distinguish knowledge traditions that originated in the Indian subcontinent	**BT 3**

Modules	Topics and Course Content	Hours
Unit 1	**Introduction to Indian Knowledge Systems (IKS):** -What is the Indian Knowledge System? -Definition of Indigenous/ Traditional Knowledge -Scope, and Importance of Traditional Knowledge. **Ancient India- Bharat Varsha:** -People of Ancient Bharat Varsha -Our great natural heritage: The great Himalayas and the rivers. - The civilizations of the Sindhu-Ganga valley, and the Brahmaputra valley. -Our coastal plains. -Our Nature: Forests and Minerals -Ancient Indian Traditional Knowledge and Wisdom about nature and climate.	15
Unit 2	**Indian Heritage of Knowledge:** -Ancient Indian Knowledge: The Vedas and its components - the Vedangas -Ancient Indian books and treaties: The Sastras. -The Great Indian Epics: The Ramayana and The Mahabharata -Epics and religious treaties of ancient Assam: Introduction to Madhav Kandali's Ramayan and Srimanta Sankardev's Dasam Skandha Bhagavat of the Puranas. -Ancient Traditional Knowledge-The Agamas -The ancient Buddhist knowledge: Tripitaka: Vinaya, Sutta and Abhidhamma Pitaka **Languages and language studies in India:** -What is linguistics? -Script and Language -Alphabet of the Indian languages Varnamala: Origin, Evolution, and phonetic features. -Languages of India -Important texts of Indian languages: Skills Siksha, Expression/Prounciation-Nirukta, Grammer-Vyakarana, Poetic rhythm Chandas. -Paninian Grammar: A Brief Introduction **Introduction to Fine Arts and Performing Arts of India:** -Ancient Indian classical music and dance forms: The Science of Dramas-Natyasastra and the Science of Music-Gandharva-Veda. -Aesthetics in Indian Art and Culture. -Folk music and traditional dance forms of the Northeast.	15
Unit 3	**Indian Science & Technology** -Ancient India's contribution to Mathematics- Number System. Algebra and Arithmetic, Geometry and Trigonometry. -Origin of Decimal system in India; nomenclature of numbers in the Vedas. Zero and Infinity.	15

	Sulba-sutras. Contribution of Brahmagupta and Sridhar Acharya to Mathematics. Important texts of Indian mathematics. • **Indian Astronomy:** Planetary System. Motion of the Planets. Velocity of Light. Eclipse. Astronomy. Navagrahas. Important works in Indian Astronomy. Aryabhata and Nilakantha: Contribution to Astronomical Studies • Indian Metal Works: Mining Techniques. Types of Metals. Tools & Techniques for Metal Smelting with examples. Metalworks in pre-modern India: Special reference to NE India	
Unit 4	**Contribution of Ancient India to Health Sciences:** - Traditional Indigenous systems of medicines in India: - Ayurveda and Yoga: Elements of Ayurveda: Gunas and Doshas, Pancha Mahabhuta and Sapta-dhatu. - Concept of disease in Ayurveda - Ayurvedic lifestyle practices: Dinacharya and Ritucharya. - Important Ayurvedic Texts - Hospitals in Ancient India • Ayurveda: Gift of India to the modern world.	15
EL	The experiential learning sessions may include: • Field Visits: Organizing visits to historical sites, museums, traditional craft centers, and other places relevant to Indian knowledge systems. • Interactive Sessions: Engaging students in discussions with experts and practitioners in various fields of Indian knowledge systems to gain insights and practical knowledge. • Online Lecture Series: Providing the students with online lectures by distinguished experts in the field of the Indian Knowledge System. • Hands-on Activities: Providing opportunities for students to participate in activities related to traditional arts, crafts, music, dance, agriculture, etc., to understand the practical aspects of • Indian knowledge systems. • Practical Demonstrations: Conducting workshops or sessions to demonstrate traditional practices, such as yoga, Ayurveda, Vastu Shastra, etc., for the students.	30
Total Credit Hours		**90**

Text Books:

1) Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavan RN. (2022), *Introduction to Indian Knowledge System: Concepts and Applications.* PHI Learning Private Ltd.
2) Mukul Chandra Bora*, Foundations of Bharatiya Knowledge System.* Khanna Book Publishing

Reference Books:

1) Baladev Upadhyaya, *Samskrta Śāstrom ka Itihās*, Chowkhambha, Varanasi, 2010.
2) D. M. Bose, S. N. Sen and B. V. Subbarayappa, Eds., *A Concise History of Science in India*, 2nd Ed., Universities Press, Hyderabad, 2010.
3) Astāngahrdaya, Vol. I, *Sūtrasthāna and Śarīrasthāna*, Translated by K. R. Srikantha Murthy, Vol. I, Krishnadas Academy, Varanasi, 1991.
4) Dharampal, *The Beautiful Tree: Indian Indigenous Education in the Eighteenth Century*, Dharampal Classics Series, Rashtrotthana Sahitya, Bengaluru, 2021.
5) J. K. Bajaj and M. D. Srinivas, *Indian Economy, and Polity in Eighteenth-century Chengalpattu*, in J. K. Bajaj ed., Indian Economy and Polity, Centre for Policy Studies, Chennai, 1995, pp. 63-84.

Type of Course: AEC	Introduction to Effective Communication Credit: 2 L-T-P-C: 1-0-0-1 Scheme of Evaluation: Theory	Course Code: CEN982A101

Course Objectives: To understand the four major aspects of communication by closely examining the processes and outlining the most effective ways to communicate with interactive activities.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	List the elements and processes that make for successful communication and recognise everyday activities that deserve closer	**BT 1**
CO 2	attention in order to improve communication skills	**BT 2**
CO 3	Contrast situations that create barriers to effective communication	**BT 3**

Modules	Topics and Course Content	Hours
Unit 1	Introduction to Effective Communication • Listening Skills ○ The Art of Listening ○ Factors that affect Listening ○ Characteristics of Effective Listening ○ Guidelines for improving Listening skills	5
Unit 2	• Speaking Skills ○ The Art of Speaking ○ Styles of Speaking ○ Guidelines for improving Speaking skills ○ Oral Communication: importance, guidelines, and barriers	5
Unit 3	• Reading Skills ○ The Art of Reading ○ Styles of Reading: skimming, surveying, scanning ○ Guidelines for developing Reading skills	5
Unit 4	• Writing Skills ○ The Art of Writing ○ Purpose and Clarity in Writing ○ Principles of Effective Writing	5
Experiential Learning: Movie/ Documentary /Podcasts screening/Peer teaching		10
Total Credit Hours		**30**

Text Books:

1) Rizvi, M. Ashraf. (2017). Effective Technical Communication. McGraw-Hill.
2) Chaturvedi, P. D. and Chaturvedi, Mukesh. (2014). Business Communication. Pearson.
3) Raman, Meenakshi and Sharma, Sangeeta. (2011). Technical Communication: Principles and Practice (2nd Edition): Oxford University Press.

Type of Course: SEC	Geological Mapping and Surveying Credit: 3 L-T-P-C: 0-0-6-3 Scheme of Evaluation: Practical	Course Code: GEOL162S111

Course Objectives: The course focuses on developing hands-on skills in collecting geological data, understanding geological formations, and creating geological maps.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Comprehend the various methods and instruments used in geological fieldwork.	**BT 1**
CO 2	Understand the significance of geological mapping and surveying in geological exploration and research.	**BT 2**
CO 3	Apply geological surveying techniques to measure and record geological features in the field.	**BT 3**
CO 4	Analyze and interpret geological field data to identify rock types, stratigraphic sequences, and structural elements.	**BT 4**
CO 5	Evaluate the reliability and accuracy of geological data collected during fieldwork.	**BT 5**
CO 6	Synthesize field data to create detailed geological maps and cross-sections.	**BT 6**

Modules	Topics and Course Content	Hours
Unit 1	Introduction to Geological Mapping and Surveying: Definition and importance of geological mapping. Overview of geological surveying techniques and instruments. Fieldwork Preparation: Planning and organizing a geological field survey. Safety considerations and fieldwork logistics.	17
Unit 2	Topographic Mapping: Understanding topographic maps and contour lines. Topographic map reading and interpretation. Concept of Toposheet indexing. Use of handheld GPS for geological surveying. Basics of Total Station Survey. Distance, height and pace approximation in geological traversing.	16
Unit 3	Topographic Map Reading (Introductory Level): Understanding contours, slope patterns, valley-ridge identification. Preparation of topographic profile from given contour map. Geological Tools Demonstration: Introduction to compass, clinometer, hand lens, and their use in simple measurement exercises.	17
Unit 4	Field visit for a period of 1 week tentatively from 1st to 7th November: Identification and classification of rocks and minerals in the field, Identification of geomorphic features, Conducting a comprehensive geological survey and mapping project, Creating a detailed geological map and report based on field data. Hands on project on Preparation of Thin sections of rocks and polished blocks.	16
Experiential Learning:		24
Total Credit Hours		**90**

Text Books suggested:

1) Guide to Field Geology – S. M. Mathur, PHI Publications
2) Field Geology – F. H. Lahee, CBS Publishers and Distributors Pvt Ltd; Sixth Edition (2002)

Reference Books:

1) Manual of Field Geology – Robert R. Compton; John Wiley & Sons.
2) Basic Methods of Structural Geology – Stephen Marshak & Gautam Mitra; Pearson Publication.

Type of Course: VAC	Earth and Climate Systems Course Level: 100 Credit: 3 L-T-P-C: 2-1-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162V101

Course Objectives: To provide a comprehensive understanding of the Earth's climate system and its interacting components, explore the natural and anthropogenic drivers of climate change over both geological and modern timescales, and analyse the impact of shifting climate patterns on global and regional environments.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Identify and list the key components of the Earth system (atmosphere, hydrosphere, cryosphere, lithosphere) and define the fundamental mechanics of Earth's solar radiation balance and albedo.	**BT 1**
CO 2	Discuss and clarify the distinctions between natural climate forcings (e.g., Milankovitch cycles, volcanic activity) and anthropogenic climate drivers, highlighting the role of major feedback loops.	**BT 2**
CO 3	Demonstrate how paleoclimatic proxy data (such as ice cores, tree rings, and marine sediments) are utilised by geologists to reconstruct historical environmental changes.	**BT 3**
CO 4	Analyse the specific impacts of modern climate change on the Indian subcontinent, critically examining variations in the Indian Summer Monsoon and rates of Himalayan glacial retreat.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	**The Earth System** Components of the Earth system. Earth's energy budget, solar radiation, albedo, and the natural greenhouse effect. Ocean-atmosphere interactions (e.g., El Niño/La Niña).	10
Unit 2	**Drivers of Climate Change** Natural forcings: Solar output variations, plate tectonics, and volcanism. Anthropogenic forcings: Greenhouse gases, aerosols, and land-use/land-cover changes. Climate feedback mechanisms.	12
Unit 3	**Earth's Climate History** Principles of palaeoclimatology. Climate proxies (ice cores, tree rings, marine/lake sediments). Major past climate events (e.g., Pleistocene Ice Ages, Holocene climatic optimum).	11
Unit 4	**Modern Impacts & Adaptation** Sea-level rise, ocean acidification, and extreme weather frequency. Regional impacts: Himalayan glacial retreat and changes in the Indian Summer Monsoon. Carbon footprint and basic mitigation strategies.	12
Experiential Learning: Home Assignments – 8 hrs, Presentation – 4 hrs, Video Screening – 4 hrs		15
Total Notional Credit Hours		**60**

Textbooks:

1) Earth's Climate: Past and Future by William F. Rudd

Reference books:

1) "The Earth System" by Lee R. Kump, James F. Kasting, and Robert G. Crane
2) "Global Physical Climatology" by Dennis L. Hartmann
3) "Paleoclimatology: Reconstructing Climates of the Quaternary" by Raymond S. Bradley.

Detailed Syllabus Of Semester 2

Type of Course: Major	Igneous Petrology and Geochemistry Course Level: 100 Credit: 4 L-T-P-C: 4-2-0-4 Scheme of Evaluation: Theory	Course Code: GEOL162M201

Course Objectives: Igneous Petrology is a course designed to provide an understanding of the origin, classification, textures, and mineralogy of igneous rocks. The course will cover the processes that lead to the formation of magmas, their emplacement and crystallization, and the resultant diversity of igneous rocks. The course will also explore the relationship between igneous processes and tectonic settings.

Geochemistry is the study of the chemical composition, structure, and processes of the Earth and other planets. This course covers the fundamentals of geochemistry, including the principles of thermodynamics, kinetics, isotope geochemistry, and major and trace element geochemistry.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Demonstrate an understanding of the fundamental principles of igneous petrology and geochemistry, including rock classification, thermodynamics, kinetics, and isotope geochemistry.	**BT 1**
CO 2	Explain the origin, evolution, and tectonic significance of magmas and interpret the role of major and trace elements in geological systems.	**BT 2**
CO 3	Apply petrographic and geochemical methods to identify, classify, and interpret igneous rocks and associated geological processes.	**BT 3**
CO 4	Analyze, interpret, and evaluate igneous and geochemical datasets using appropriate analytical, statistical, and graphical techniques for mineral exploration and geological investigations.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	Introduction: Heat flow, geothermal gradient, Physical and chemical properties of magmas. Classification and nomenclature of igneous rocks. Textures and structures of igneous rocks. Mode of occurrence of Igneous rocks. Types of magma sources. Magma chambers. Melting processes in the Earth's mantle and crust. Magma ascent and eruption and their products. Nucleation and Crystallisation rates. Crystallisation of Magma, Reaction Principle, Mechanisms of Magmatic differentiation, Role of volatiles in magmatic differentiation.	15
Unit 2	Classification schemes for plutonic and volcanic rocks. Petrogenesis of Felsic and Mafic igneous rocks: Granitoids, Basalt, Gabbro, Alkaline rocks, peridotites and kimberlites. Continental rifting and flood basalts. Plate tectonics and igneous rock formation. Igneous rocks as indicators of tectonic processes. Oceanic crust formation and composition. Volcanic eruptions and associated rocks. Volcanic hazards monitoring, Barren Island Volcanics. Igneous rocks of NE India.	15
Unit 3	Definition and scope of geochemistry in Earth sciences. Major geological reservoirs and elemental abundances in the Earth's crust. Introduction to properties of elements: The periodic table, atomic environment of elements. Geochemical classification of elements. Types of chemical bonds in minerals and rocks. Crystal structures and their influence on mineral properties. Geochemical cycles of major elements (C, O, N, S) in the Earth's crust. Concepts of mass balance. Chemical reactions and equations. Conservation of mass, isotopic and elemental fractionation. Fractionation and partitioning of elements during geological processes.	15

Unit 4	Chemical differentiation and Composition of the Earth (Continental crust, Oceanic crust, depleted mantle, enriched mantle and core). Geology of Meteorites. Cosmic abundance of elements. Geochemical variability of magma and its products. Basic concepts of Aqueous geochemistry, Eh-pH relations. Mineral reactions- diagenesis and hydrothermal reactions. Chemical changes during metamorphism and metasomatism. Stable and radiogenic isotope systems and their applications. Isotopic dating methods and their use in geochronology.	15
Experiential Learning: Home Assignments – 8 hrs, Presentation – 4 hrs, Video Screening – 4 hrs		16
Total Notional Credit Hours		**60**

Text Books Suggested:

1) Principles of igneous and metamorphic petrology – A. Philpotts & J. Ague. (2009). Cambridge University Press.
2) Principles of igneous and metamorphic petrology – J. D. Winter (2014). Pearson
2) Geochemistry – W. M. White, (2013), Wiley-Blackwell Publishing.
3) Introduction to Geochemistry: Principles and Applications - Kula C. Misra, (2012), Wiley-Blackwell Publishing.

Reference Books:

1) Petrology: the study of igneous, sedimentary, and metamorphic rocks. L. A. Raymond, (2002). McGraw-Hill Science Engineering
2) Principles of Petrology – G. W. Tyrrell. (1926). Springer
3) Principles of Geochemistry - Mason, B., (3rd Edition, 1986), Wiley New York.
4) Essentials of geochemistry - Walther, J. V. (2009), Jones & Bartlett Publishers.
5) Geochemistry: An introduction - Francis Albarède, Cambridge University Press, 2009.

Type of Course: Major	Igneous Petrology and Geochemistry Practical Course Level: 100 Credit: 2 L-T-P-C: 0-1-4-2 Scheme of Evaluation: Practical	Course Code: GEOL162M212

Course Objectives: Igneous Petrology is a course designed to provide an understanding of the origin, classification, textures, and mineralogy of igneous rocks. The course will cover the processes that lead to the formation of magmas, their emplacement and crystallization, and the resultant diversity of igneous rocks. The course will also explore the relationship between igneous processes and tectonic settings.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Classify igneous rocks based on their mineralogy and textures.	**BT 1**
CO 2	Describe the processes involved in the formation of magmas and their subsequent crystallization into igneous rocks. Explain the relationship between igneous processes and tectonic settings.	**BT 2**
CO 3	Analyse and interpret igneous rock suites using microscopic and macroscopic techniques.	**BT 3**
CO 4	Evaluate the applications of igneous petrology in geologic exploration and mineral resource identification.	**BT 4**

Modules	Topics and Course Content	Hours
I	**Megascopic Identification of Igneous Rocks:** Description and classification based on texture, mineralogy, and colour. **Textural Study in Hand Specimens:** Granular, porphyritic, vesicular, glassy, intergranular, ophitic, poikilitic textures. **Classification of Igneous Rocks:** Based on mineral composition (QAPF diagram) and silica content (acidic-basic). **Petrographic Study of Igneous Rocks in Thin Sections:** Basalt, andesite, granite, diorite, syenite, anorthosites, peridotite, gabbro, rhyolite, dacite. **Textural Interpretation from Thin Sections:** Inference of cooling history and crystallization sequence. **Calculation of CIPW Norms:** Introduction and hands-on calculation using oxide data. **Conversion of Oxide Percentages to Molecular Proportions:** Calculation of cation proportions and structural formulae of minerals. **Geochemical Variation Diagrams:** Harker diagrams, AFM diagrams, and other binary plots. **Mass Balance and Mixing Calculations:** Basic exercises on fractional crystallization and magma mixing. **Field visit to observe the megascopic characteristics of the igneous rocks.**	30
Total Notional Credit Hours		**30**

Text Books Suggested:

1) Principles of igneous and metamorphic petrology – A. Philpotts & J. Ague. (2009). Cambridge University Press.
2) Principles of igneous and metamorphic petrology – J. D. Winter (2014). Pearson

Reference Books:

1) Petrology: the study of igneous, sedimentary, and metamorphic rocks. L. A. Raymond, (2002). McGraw-Hill Science Engineering
2) Principles of Petrology – G. W. Tyrrell. (1926). Springer

Type of Course: Minor	Geology and Natural Hazards Course Level: 100 Credit: 3 L-T-P-C: 3-1-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162N201

Course Objectives: The course explores the geological factors and interactions that lead to earthquakes, volcanic eruptions, landslides, floods, and other geological hazards. It also emphasizes the understanding of hazard assessment, mitigation, and their impact on society and the environment.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Remember and identify the geological and environmental impacts of different hazards.	**BT 1**
CO 2	Understand the geological processes responsible for natural hazards.	**BT 2**
CO 3	Apply geological and geophysical principles to hazard assessment and mapping.	**BT 3**
CO 4	Analyse case studies of past geological hazards and their societal implications.	**BT 4**
CO 5	Evaluate the role of geologists in assessing and mitigating natural hazards.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	Definition and classification of natural hazards Overview of geological processes and their relation to hazards Plate tectonics and seismicity, Seismic waves and earthquake mechanisms Volcanic processes and types of eruptions Volcanic landforms and volcanic risk assessment	17
Unit 2	Causes and triggers of landslides Landslide types and susceptibility mapping Floods and its contributing factors, Concept of Flood cyclicity Floodplain and flood risk assessment	16
Unit 3	Tsunami generation and propagation Coastal erosion and impact on communities Glacial Lake Outburst Flood Sink-holes – its causes and environmental impact. Radon emanation and its potential health hazards.	17
Unit 4	Development of hazard mitigation plans and policies Analysis of historic geological disasters and their impact Social and environmental justice in hazard response Ethical responsibilities of geologists in hazard assessment and communication	16
Experiential Learning: Home Assignments – 10 hrs, Presentation – 10 hrs, Video Screening – 4 hrs		24
Total Notional Credit Hours		**90**

Text Books Suggested:

1) Natural Hazards and Disasters: Donald Hyndman, David Hyndman, 5th Edition, 2020, Cengage Learning.
2) Earthquakes and Geological Hazards - The Next Generation: Timothy L. Hall, Michael L. Anderson, 2nd Edition, 2021, Wiley-Blackwell.

Reference Books:

1) Volcanoes – Global Perspectives: John P. Lockwood, Richard W. Hazlett, 3rd Edition, 2015, Wiley-Blackwell.
2) Landslides – Types, Mechanisms, and Modelling: Jean Hutchinson, 1st Edition, 2011, Cambridge University Press.
3) Atmosphere, Clouds, and Climate: David Randall, 1st Edition, 2012, Princeton University Press.

Type of Course: IDC	Introduction to Indian Knowledge System - II Credit: 3 L-T-P-C: 2-1-0-3 Scheme of Evaluation: Theory	Course Code: IKS992K201

Course Objectives: This Foundation course is designed to present an overall introduction to all the streams of IKS relevant to the UG program. It would enable students to explore the most fundamental ideas that have shaped Indian Knowledge Traditions over the centuries.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall traditional Indian knowledge traditions constituting Indian culture.	**BT 1**
CO 2	Summarize differences between classical literature in Sanskrit and other Indian languages.	**BT 2**
CO 3	Compare knowledge traditions originating in NE India.	**BT 3**
CO 4	Appreciate the contribution of Indian Knowledge Systems to the World.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	**Indian Classical Literature** Indian Classical Literature: A Brief Introduction. - Ancient Indian Spritual Poetics-Kavya: Contribution of Kalidasa **Diversity and Indian Culture:** - Diversity and Indian Culture - Indigenous Faith and Religion - Preservation of culture and indigenous knowledge **The Purpose of Knowledge** - Understanding Self-Awareness and Spirituality. - Indian concept and purpose of Knowledge and Education - Understanding Spirituality and Materialism: Para and Apara Vidya	15
Unit 2	**Methodology of Indian Knowledge System:** - Shruti and Smriti traditions. - Introduction to Shastras. - Manuscriptology: The art and science of documenting knowledge. - Repositories of ancient manuscripts with special reference to the Northeast India. **Indian Architecture and Town Planning:** - Introduction ancient Indian architecture. - Sthapatya-Veda: An Introduction - Indigenous tools & techniques for town planning & Temple Architecture. Lothal, Mohan Jo Daro. - Temple Art: Lepakshi Temple, Jagannath Puri Temple, Konark Sun Temple. - Vernacular architecture of Assam: Special reference to Brahmaputra Valley	15
Unit 3	**Indian Agriculture:** - Agriculture: Significance in Human Civilization. - Sustainable Agriculture. - Historical significance of agriculture and sustainable farming in India. - Step Cultivation of India: Special reference to Northeast India. - Wet rice cultivation of Assam. **Indian Textiles:** - What is Textile? - Tradition of cotton and silk textiles in India. - The historical contribution of textile and weaving to the Indian economy. - Varieties of textiles and dyes developed in different regions of India with special reference to Northeast India	15

Unit 4	**Indian Polity and Economy:** - Understanding Kingdom and Chiefdom - Role of a king - The Indian idea of a well-organized polity and flourishing economy. - The Chakravarti System: Administrative System of Ancient Bharatvarsha. - Village administrative system: Northeast India. - Arthashastra: Brief synopsis **The outreach of Indian Knowledge System across Geographical Boundaries** - Indian Languages. - Scripts. - Linguistics. - Ayurveda. - Yoga and Meditation. - Textile - Decimal value place system-based arithmetic, Algebra and Astronomy	15
EL	The experiential learning sessions may include: • Field Visits: Organizing visits to historical sites, museums, traditional craft centers, and other places relevant to Indian knowledge systems. • Interactive Sessions: Engaging students in discussions with experts and practitioners in various fields of Indian knowledge systems to gain insights and practical knowledge. • Online Lecture Series: Providing the students with online lectures by distinguished experts in the field of the Indian Knowledge System. • Hands-on Activities: Providing opportunities for students to participate in activities related to traditional arts, crafts, music, dance, agriculture, etc., to understand the practical aspects of • Indian knowledge systems. • Practical Demonstrations: Conducting workshops or sessions to demonstrate traditional practices, such as yoga, Ayurveda, Vastu Shastra, etc., for the students.	30
Total Notional Credit Hours		**90**

Text Books:

1) Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavan RN. (2022), *Introduction to Indian Knowledge System: Concepts and Applications.* PHI Learning Private Ltd.
2) Mukul Chandra Bora*, Foundations of Bharatiya Knowledge System.* Khanna Book Publishing

Reference Books:

1) Baladev Upadhyaya, *Samskrta Śāstrom ka Itihās*, Chowkhambha, Varanasi, 2010.
2) D. M. Bose, S. N. Sen and B. V. Subbarayappa, Eds., *A Concise History of Science in India*, 2nd Ed., Universities Press, Hyderabad, 2010.
3) Astāngahrdaya, Vol. I, *Sūtrasthāna and Śarīrasthāna*, Translated by K. R. Srikantha Murthy, Vol. I, Krishnadas Academy, Varanasi, 1991.
4) Dharampal, *The Beautiful Tree: Indian Indigenous Education in the Eighteenth Century*, Dharampal Classics Series, Rashtrotthana Sahitya, Bengaluru, 2021.
5) J. K. Bajaj and M. D. Srinivas, *Indian Economy, and Polity in Eighteenth-century Chengalpattu*, in J. K. Bajaj ed., Indian Economy and Polity, Centre for Policy Studies, Chennai, 1995, pp. 63-84.

Type of Course: AEC	Approaches to Verbal and Non-Verbal Communication Credit: 1 L-T-P-C: 1-0-0-1 Scheme of Evaluation: Theory	Course Code: CEN982A201

Course Objectives: To introduce the students to the various forms of technical communication and enhance their knowledge in the application of both verbal and non-verbal skills in communicative processes.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	List the different types of technical communication, their characteristics, their advantages and disadvantages.	**BT 1**
CO 2	Explain the barriers to communication and ways to overcome them.	**BT 2**
CO 3	Identify the means to enhance conversation skills.	**BT 3**
CO 4	Determine the different types of non-verbal communication and their significance.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	**Technology Enabled Communication** Communicating about technical or specialized topics, Different forms of technology-enabled communication tools used in organizations Telephone, Teleconferencing, Fax, Email, Instant messaging, Blog, Podcast, Videos, videoconferencing, social media	4
Unit 2	**Communication Barriers** Types of barriers: Semantic, Psychological, Organisational, Cultural, Physical, Physiological, Methods to overcome barriers to communication.	4
Unit 3	**Conversation skills/Verbal Communication** Conversation – Types of Conversation, Strategies for Effectiveness, Conversation Practice, Persuasive Functions in Conversation, Telephonic Conversation and Etiquette Dialogue Writing, Conversation Control.	4
Unit 4	**Non-verbal Communication** Body language- Personal Appearance, Postures, Gestures, Eye Contact, Facial expressions Paralinguistic Features-Rate, Pause, Volume, Pitch/Intonation/ Voice/Modulation, Proxemics, Haptics, Artifacts, Chronemics,	4
Experiential Learning: Movie/ Documentary /Podcasts screening/Peer teaching		10
Total Notional Credit Hours		**26**

Text Books:

1) Rizvi, M. Ashraf. (2017). Effective Technical Communication. McGraw-Hill.
2) Chaturvedi, P. D. and Chaturvedi, Mukesh. (2014). Business Communication. Pearson.
3) Raman, Meenakshi and Sharma, Sangeeta. (2011). Technical Communication: Principles and Practice (2nd Edition): Oxford University Press.

Type of Course: SEC	Geomorphology and Landform Analysis Credit: 3 L-T-P-C: 0-1-6-3 Scheme of Evaluation: Practical	Course Code: GEOL162S211

Course Objectives: This practical and field-based course focuses on developing students' practical skills in understanding and interpreting various landforms and their significance in geological studies.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Identify different landforms and understand their formation mechanisms.	**BT 1**
CO 2	Comprehend the relationship between tectonics, climate, and surface processes in shaping landforms.	**BT 2**
CO 3	Apply geomorphological principles to interpret landscape evolution during field studies.	**BT 3**
CO 4	Analyse the formation and significance of specific landforms through field investigations.	**BT 4**
CO 5	Evaluate the influence of human activities on landform evolution through practical case studies.	**BT 5**
CO 6	Integrate geomorphological knowledge with geological mapping during field exercises.	**BT 6**

Modules	Topics and Course Content	Hours
Unit 1	Introduction to field instruments and geospatial tools Field study of river systems and their dynamics Field identification and interpretation of erosional and depositional features	20
Unit 2	Study of river channel, levee and floodplains, characteristics in a particular location (example Chandrapur, Morigaon, Palashbari etc.) Practical exercises on sediment characterization and analysis Understanding sedimentary structures and their significance	20
Unit 3	Palaeocurrent analysis in fluvial systems (Rose diagrams, Moving vector analysis). Evaluation of river bank erosion and deposition. Analysis of anthropogenic activities on levees and flood plains. Plane table surveying, Dumpy level, Total Station.	25
Unit 4	Practical application of GIS, remote sensing, and digital terrain models for fluvial mapping. Preparation of simple lithostratigraphic columns of river banks. Preparation of river bank profiles. Study of fluvial landforms using geospatial tools Identification and analysis of important geomorphic features in the field: valleys, gullies, river terraces, river meanders, floodplains, levee, point bars, paleochannels. Field study report.	25
Total Credit Hours		**90**

Text Books suggested:

1) Fluvial Geomorphology: Luna B. Leopold, M. Gordon Wolman, and John P. Miller, 1st Edition, 2014, W. H. Freeman and Company.
2) Geomorphology and Global Environmental Change: Olav Slaymaker and Thomas Spencer, 1st Edition, 2012, Cambridge University Press.

Reference Books:

1) Principles of Geomorphology: William D. Thornbury, 1st Edition, 2019, Wiley-Blackwell.
2) Applied Fluvial Geomorphology for River Engineering and Management: Philip J. Ashworth, Gary J. Brierley, and G. Mathias Kondolf, 2nd Edition, 2019, Wiley-Blackwell.

Type of Course: VAC	Disaster Management in the Indian Context Course Level: 100 Credit: 3 L-T-P-C: 2-1-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162V201

Course Objectives: To introduce the fundamental concepts of hazards, vulnerabilities, and disasters, to understand India's specific vulnerability profile with a special emphasis on the seismic and hydro-meteorological hazards of Northeast India, and familiarise students with the institutional frameworks for disaster management in India.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall and define core disaster management concepts, including hazard, vulnerability, risk, and capacity, while listing the major classifications of natural and man-made disasters.	**BT 1**
CO 2	Explain the roles, responsibilities, and structural hierarchy of India's institutional framework (NDMA, SDMA, DDMA) as mandated by the Disaster Management Act 2005.	**BT 2**
CO 3	Apply the principles of the Disaster Management Cycle to develop basic, actionable community-based preparedness plans and emergency response scenarios for local institutions.	**BT 3**
CO 4	Analyse localised disaster case studies (such as the Brahmaputra floods or regional Northeast landslides) to evaluate the effectiveness of structural versus non-structural mitigation strategies.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	**Introduction to Disasters** Concept of hazard, vulnerability, risk, and capacity. Classification of natural and man-made disasters. The Disaster Management Cycle (Preparedness, Response, Recovery, Mitigation).	10
Unit 2	**India's Disaster Profile** Seismicity of India and seismic zonation. Hydro-meteorological disasters: Floods, cyclones, and droughts. Focus on the Northeast: Landslides, riverbank erosion, and flash floods.	12
Unit 3	**Institutional Framework in India** Disaster Management Act 2005. Structure and functions of NDMA, SDMA, and DDMA. Role of NDRF and SDRF in emergency response.	12
Unit 4	**Mitigation and Community Action** Structural and non-structural mitigation measures. Early warning systems. Community-based disaster risk reduction (CBDRR) and the role of youth/students in disaster management.	11
Experiential Learning: Home Assignments – 8 hrs, Presentation – 4 hrs, Video Screening – 4 hrs		15
Total Notional Credit Hours		**60**

Textbooks:

1) Disaster Management, Gupta, A.K. & Goyal, P.K. (2026). Khanna Publishing House, New Delhi.

Reference books:

1) Disaster Management in India: Policies, Institutions, Practices, Pandey, K. (2025). Routledge India.
2) Disaster Management: A Comprehensive Approach; Author: S. Lakshmi
3) National Disaster Management Authority (NDMA) Publications: National Policy on Disaster Management (2009)

Detailed Syllabus Of Semester 3

Type of Course: Major	Metamorphic Petrology			Course Code:
	Course Level: 200	Credit: 3	L-T-P-C: 3-0-0-3	GEOL162M301
		Scheme of Evaluation: Theory		

Course Objectives: This course provides an in-depth understanding of the origin, classification, and petrological properties of metamorphic rocks. Students will learn about the various metamorphic processes, including the role of fluids and deformation, and how these processes influence mineral assemblages and textures. The course also covers the use of metamorphic petrology in understanding the tectonic and thermal history of a region.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Describe the classification and nomenclature of metamorphic rocks.	**BT 1**
CO 2	Identify and interpret the mineral assemblages and textures of metamorphic rocks, and explain their significance.	**BT 2**
CO 3	Apply knowledge of metamorphic processes and petrographic techniques to identify and interpret metamorphic rocks and their evolution.	**BT 3**
CO 4	Analyse the factors that control the metamorphic process, including pressure, temperature, fluids, and deformation.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	**Metamorphism:** Definition and limits of metamorphism. Factors controlling metamorphism: temperature, pressure, chemically active fluids, and time. Types of metamorphism - contact metamorphism, regional metamorphism, fault zone metamorphism, impact metamorphism.	9
Unit 2	**Metamorphic facies and grades** Mineralogical phase rule of closed and open system. Index minerals and metamorphic zones, Isograde, and metamorphic grades Chemographic projections. Concept of metamorphic facies and facies series	9
Unit 3	**Metamorphism and Tectonism & Petrogenesis** Relationship between metamorphism and deformation. Structure and textures of metamorphic rocks. Metamorphic mineral reactions (prograde and retrograde).	9
Unit 4	**Petrogenesis** Migmatites and their origin. Metasomatism and role of fluids in metamorphism. Petrogenesis of important metamorphic rock associations- schists, gneisses, khondalites, charnockites, blueschists and eclogites.	9
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Credit Hours		**90**

Text books:

1) Principles of igneous and metamorphic petrology – A. Philpotts & J. Ague. (2009). Cambridge University Press.
2) Principles of igneous and metamorphic petrology – J. D. Winter (2014). Pearson

Reference books:

1) Petrology: the study of igneous, sedimentary, and metamorphic rocks. L. A. Raymond, (2002). McGraw-Hill Science Engineering
2) Igneous and Metamorphic Petrology - Myron G. Best (2001).

Type of Course: Major	Metamorphic Petrology Practical			Course Code:
	Course Level: 200	Credit: 1	L-T-P-C: 0-0-2-1	GEOL162M312
		Scheme of Evaluation: Practical		

Course Objectives: This course provides an in-depth understanding of the origin, classification, and petrological properties of metamorphic rocks. Students will learn about the various metamorphic processes, including the role of fluids and deformation, and how these processes influence mineral assemblages and textures. The course also covers the use of metamorphic petrology in understanding the tectonic and thermal history of a region.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Describe the classification and nomenclature of metamorphic rocks.	**BT 1**
CO 2	Identify and interpret the mineral assemblages and textures of metamorphic rocks, and explain their significance.	**BT 2**
CO 3	Apply knowledge of metamorphic processes and petrographic techniques to identify and interpret metamorphic rocks and their evolution.	**BT 3**
CO 4	Analyse the factors that control the metamorphic process, including pressure, temperature, fluids, and deformation.	**BT 4**

Modules	Topics and Course Content	Hours
I	**Hand Specimens study of Megascopic characteristics and Metamorphic structures of common Metamorphic Rocks:** slate, phyllite, schist, gneiss, marble, quartzite and migmatite. **Petrographic study of Metamorphic Rocks and recognition of Metamorphic textures (**Foliation, lineation, crenulation, granoblastic texture, idioblastic boundaries**) under microscope:** quartzite, marble, chlorite schist, biotite schist, garnet schist, gneiss, charnockite. **Graphical Representation of Mineral Assemblages:** Chemographic diagrams (e.g., ACF, AKF diagrams) and plotting common assemblages. **Metamorphic Zones and Isograds (Map Exercise):** Interpretation of metamorphic zones using index minerals and drawing isograds. **Study of Indian Rock Types:** Charnockite, Khondalite, Gondite, kodurite	30
Total Credit Hours		**30**

Text books:

1) Principles of igneous and metamorphic petrology – A. Philpotts & J. Ague. (2009). Cambridge University Press.
2) Principles of igneous and metamorphic petrology – J. D. Winter (2014). Pearson

Reference books:

1) Petrology: the study of igneous, sedimentary, and metamorphic rocks. L. A. Raymond, (2002). McGraw-Hill Science Engineering
2) Igneous and Metamorphic Petrology - Myron G. Best (2001).

Type of Course: Major	Sedimentology			Course Code:
	Course Level: 200	Credit: 3	L-T-P-C: 3-0-0-3	GEOL162M303
		Scheme of Evaluation: Theory		

Course Objectives: This course will focus on the processes of sedimentation, the diagenesis of sediments, and the properties and classification of sedimentary rocks. Topics covered will include sedimentary environments, depositional processes, sedimentary structures, mineralogy, texture, and sedimentary rock classification. The course will also cover the interpretation of sedimentary rocks in terms of paleoenvironmental and paleoclimatic conditions.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Students will be able to recall and recognize the key concepts, principles, and facts related to sedimentary petrology.	**BT 1**
CO 2	Students will be able to explain the processes of sedimentation, diagenesis, and lithification that result in the formation of sedimentary rocks.	**BT 2**
CO 3	Students will be able to apply the principles and concepts of sedimentary petrology to analyze and interpret the origin, composition, and classification of sedimentary rocks.	**BT 3**
CO 4	Students will be able to analyse sedimentary rocks in terms of their texture, mineralogy, sedimentary structures, and depositional environments.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	Weathering and Erosion: Physical, chemical and biological weathering. Sedimentary texture: size, shape, roundness, sphericity, fabric, packing. Concepts of diagenesis, Stages of diagenesis, Compaction and cementation.	9
Unit 2	Textural classification of sediments and sedimentary rocks. Sediment dynamics: Nature of fluid flow – Laminar vs. turbulent flow, concept of flow regime and sediment transport. Sedimentary structures – bedforms and internal stratification. Brief description of breccia, Conglomerates, Sandstones, Shales, Mudstones, Limestones	9
Unit 3	Concept of sedimentary facies. Facies sequences in fluvial, marine, desert, glacial and lacustrine environments	9
Unit 4	Concept of Paleocurrent analysis. Introduction to Mineralogical classification of sediments and sedimentary rocks (clastics and non-clastics). Geochemical fence.	9
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Credit Hours		**90**

Text Books Suggested:

1) Introduction to Sedimentology – S. M. Sengupta, (2018), CBS.
2) Depositional Sedimentary Environments - Reineck and Singh, (1980), Springer – Verlag.

Reference Books:

1) Sedimentology and Stratigraphy - Nichols, G. (2009), Second Edition. Wiley Blackwell.
2) Sedimentary Rocks – F. J. Pettijohn.
3) Sedimentary Petrology – Tucker, M. E. (2006), Blackwell Publishing.
4) Petrology of Sedimentary Rocks – Sam Boggs, (2nd edition, 2009), Cambridge University Press, New York.

Type of Course: Major	Sedimentology Practical - I	Course Code:
	Course Level: 200 Credit: 1 L-T-P-C: 0-0-2-1 Scheme of Evaluation: Practical	GEOL162M314

Course Objectives: This course will focus on the processes of sedimentation, the diagenesis of sediments, and the properties and classification of sedimentary rocks. Topics covered will include sedimentary environments, depositional processes, sedimentary structures, mineralogy, texture, and sedimentary rock classification. The course will also cover the interpretation of sedimentary rocks in terms of paleoenvironmental and paleoclimatic conditions.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Students will be able to recall and recognize the key concepts, principles, and facts related to sedimentary petrology.	BT 1
CO 2	Students will be able to explain the processes of sedimentation, diagenesis, and lithification that result in the formation of sedimentary rocks.	BT 2
CO 3	Students will be able to apply the principles and concepts of sedimentary petrology to analyze and interpret the origin, composition, and classification of sedimentary rocks.	BT 3
CO 4	Students will be able to analyse sedimentary rocks in terms of their texture, mineralogy, sedimentary structures, and depositional environments.	BT 4

Modules	Topics and Course Content	Hours
I	**Megascopic Identification of Common Sedimentary Rocks:** sandstone, shale, limestone, conglomerate, breccia, etc. **Grain Size Analysis of clastic sediments (Sieve and Pipette Method),** Graphical representation of grain size distribution using histograms and cumulative curves. **Determination of Statistical Parameters** using graphical methods (Folk & Ward method) and their interpretation. **Determination of Grain Shape:** Roundness, and Sphericity using Riley's method and Wadell method **Study of Sedimentary Structures in Hand Specimens and in field:** Identification and interpretation of bedding, cross-bedding, graded bedding, ripple marks, mud cracks, etc. **Paleocurrent Analysis:** Measurement of current directions from ripple marks or cross-bedding; rose diagram plotting and interpretation. **Microscopic Study of Clastic and Non clastic Sedimentary Rocks:** Thin section identification of minerals, matrix, cements, and textures in sandstone, shale, and limestone. **Mineralogical (Dunham, Folk, 1964) Classification of Sedimentary Rocks** **Facies Interpretation Exercise:** Identification of sedimentary facies and facies association in field. (Field visit for a period of 1 week tentatively from 1st to 7th November)	30
Total Credit Hours		**30**

Text Books Suggested:

1) Introduction to Sedimentology – S. M. Sengupta, (2018), CBS.
2) Depositional Sedimentary Environments - Reineck and Singh, (1980), Springer – Verlag.

Reference Books:

1) Sedimentology and Stratigraphy - Nichols, G. (2009), Second Edition. Wiley Blackwell.
2) Sedimentary Rocks – F. J. Pettijohn.
3) Petrology of Sedimentary Rocks – Sam Boggs, (2nd edition, 2009), Cambridge University Press, New York.

Type of Course: Minor	Mineralogy Course Level: 200 Credit: 4 L-T-P-C: 3-1-0-4 Scheme of Evaluation: Theory	Course Code: GEOL162N301

Course Objectives: Mineralogy is an undergraduate-level course that focuses on the study of minerals and their properties, including crystallography, optical properties, chemical composition, and physical characteristics. The course provides an overview of the formation and classification of minerals, as well as the processes involved in their identification and analysis.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Remember and identify the properties of minerals, their classification, and crystallographic systems.	**BT 1**
CO 2	Explain the physical and chemical properties of minerals and their economic importance.	**BT 2**
CO 3	Apply mineralogical knowledge to identify minerals using optical microscopy and physical characteristics of the specimens.	**BT 3**
CO 4	Analyse various crystallographic and mineralogical data to identify and classify them into various classes.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	Concept of crystal, crystalline and amorphous substances. Concept of Unit cell and Lattice structures, Bravais Lattices; Types of crystal structures (e.g., cubic, hexagonal); Symmetry elements and point groups; Crystallographic axes and planes; Overview of crystal systems (e.g., isometric, tetragonal, etc.); Crystallographic axes and symmetry elements for each system.	9
Unit 2	Interfacial angle, crystal parameters and indices. Stereograms and Hermann- Mauguin System. Relationship between crystallography and mineral properties. Minerals - definition, physical and chemical properties; Chemical classification of minerals.	9
Unit 3	Silicate and non-silicate structures of minerals. Study of physical properties of minerals of the following group of minerals: Silica and Feldspar and Mica.	9
Unit 4	Polarization of light, Polarisers. Functions of petrological microscope. Optical behaviour of minerals: Absorption, Transmission and Double-refraction of light. Theory of light propagation in minerals: Isotropy and Anisotropy; Optical properties of minerals in thin section.	9
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Credit Hours		**90**

Text Books Suggested:

1) Mineralogy - Dexter Perkins, 3rd edition (2015), Pearson Publication.

Reference Books:

1) Introduction to Optical Mineralogy – William D. Nesse, 3rd edition (2004), Oxford University Press.
2) The Manual of Mineral Science (after James D. Dana) - Dutrow, B., Dwight, J., & Klein, C.; (2007) J. Wiley & Sons.
3) Introduction to Rock Forming Minerals – W. A. Deer, R. A. Howie, and J. Zussman, 3rd edition (2013), Prentice Hall

Type of Course: IDC	PHYSICS AND CHEMISTRY OF EARTH Credit: 3 L-T-P-C: 3-0-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162I301

Course Objectives: The course provides an integrated understanding of Earth's physical features, interior structure, and magnetism. It covers the geochemical processes that determine the origin and abundance of elements, as well as isotope applications in geological processes. Additionally, the course explores mantle convection and phase transitions, equipping students with the knowledge to analyse and model the interplay between these geological and geophysical phenomena.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	List and describe the Earth's major surface features and internal structure, including key sub-divisions and discontinuities.	**BT 1**
CO 2	Explain the principles of Earth's magnetism and the processes driving magnetic field production and secular variation.	**BT 2**
CO 3	Apply geochemical classification principles to classify elements based on their abundance and characteristics within the Earth and the solar system.	**BT 3**
CO 4	Analyse the processes of Earth accretion and early differentiation to interpret the planet's current geochemical and structural framework.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	**Earth's Physical Features and Interior** Earth's surface features: continents, continental margins, oceans. Earth's interior: variation of physical quantities and seismic wave velocity inside the Earth.	10
Unit 2	**Earth Interior and Dynamics** Major sub-divisions and discontinuities of the Earth's interior. Core: seismological and other geophysical constraints. Convection in the mantle.	10
Unit 3	**Earth's Magnetism and Core Dynamics** Elements of Earth's magnetism. Convections in the Earth's core and production of magnetic field. Secular variation and westward drift. Solar activity and magnetic disturbance.	12
Unit 4	**Geochemistry and Elemental Composition** Elements: origin of elements/nucleosynthesis. Abundance of elements in the solar system/planet Earth. Geochemical classification of elements. Earth accretion and early differentiation.	13
Experiential Learning: Home Assignments – 15 hrs, Presentation – 15 hrs, Video Screening – 15 hrs		45
Total Credit Hours		**90**

Text Books Suggested:

1) Essentials of Geology - Stephen Marshak, 4th edition, W. W. Norton & Company
2) Introduction to Geochemistry: Principles and Applications - Kula C. Misra, (2012), Wiley-Blackwell Publishing.

Reference Books:

1) Essentials of geochemistry - Walther, J. V. (2009), Jones & Bartlett Publishers.
2) Introduction to Physical Geology – Thompson & Turk

Type of Course: AEC	Fundamentals of Business Communication Credit: 2 L-T-P-C: 1-1-0-2 Scheme of Evaluation: Theory	Course Code: CEN982A301

Course Objectives: The aim of the course is to develop essential business communication skills, including effective writing, speaking, and interpersonal communication, to enhance professional interactions, collaboration, and successful communication strategies within diverse corporate environments.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define and list business documents using appropriate formats and styles, demonstrating proficiency in written communication for various business contexts.	**BT 1**
CO 2	Demonstrate confident verbal communication skills through persuasive presentations, active listening, and clear articulation to engage and influence diverse stakeholders.	**BT 2**
CO 3	Apply effective interpersonal communication strategies, including conflict resolution and active teamwork, to foster positive relationships and contribute to successful organizational communication dynamics	**BT 3**

Modules	Topics and Course Content	Hours
Unit 1	**Business Communication: Spoken and Written** • The Role of Business Communication • Classification and Purpose of Business Communication • The Importance of Communication in Management • Communication Training for Managers • Communication Structures in Organizations • Information to be Communicated at the Workplace • Writing Business Letters, Notice, Agenda and Minutes	5
Unit 2	**Negotiation Skills in Business Communication** • The Nature and Need for Negotiation ○ Situations requiring and not requiring negotiations • Factors Affecting Negotiation ○ Location, Timing, Subjective Factors • Stages in the Negotiation Process ○ Preparation, Negotiation, Implementation • Negotiation Strategies	5
Unit 3	**Ethics in Business Communication** • Ethical Communication • Values, Ethics and Communication • Ethical Dilemmas Facing Managers • A Strategic Approach to Business Ethics • Ethical Communication on the Internet • Ethics in Advertising	5
Unit 4	**Business Etiquettes and Professionalism** • Introduction to Business Etiquette • Interview Etiquette • Social Etiquette • Workplace Etiquette • Netiquette	5
Total Credit Hours		**20**

Text Books:

1) Business Communication by Shalini Verma
2) Business Communication by P.D. Chaturvedi and Mukesh Chaturvedi
3) Technical Communication by Meenakshi Raman and Sangeeta Sharma

Type of Course: SEC	Remote Sensing and GIS Credit: 3 L-T-P-C: 0-0-6-3 Scheme of Evaluation: Practical	Course Code: GEOL162S311

Course Objectives: This course introduces the fundamental principles of remote sensing and Geographic Information Systems (GIS) and their applications in Earth Sciences. The course covers the principles of electromagnetic radiation, remote sensing sensors, and image interpretation techniques. Students will also learn the basic concepts of GIS and spatial analysis.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall basic concepts and facts related to remote sensing and GIS.	**BT 1**
CO 2	Explain the principles and theories behind remote sensing and GIS techniques.	**BT 2**
CO 3	Apply remote sensing and GIS techniques to analyse and interpret spatial data.	**BT 3**
CO 4	Analyse and interpret remotely sensed data to derive meaningful information.	**BT 4**
CO 5	Synthesize their knowledge by mastering thematic mapping and symbology in geology.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Fundamentals of Geospatial Technologies & Software Orientation** Basics of Remote Sensing: EMR, types of resolutions, sensors, and platforms. Introduction to GIS: Spatial vs non-spatial data, raster vs vector, layers and formats. Coordinate systems: Geographic vs Projected, UTM, EPSG codes. Introduction to GIS: Interface, settings, loading base maps. Exploring GIS user interface. Loading vector and raster datasets. Coordinate reference system assignment and conversion. Downloading and exploring satellite images (from USGS/ISRO portals).	16
Unit 2	**Map Handling, Editing & Georeferencing** Concept of georeferencing: image-to-map transformation. Topographic maps and geological maps: scales, symbols, legends. Georeferencing a scanned topographic/geological map. Creating and editing shapefiles. Creating vector data layers: point, line, polygon features. Understanding attribute tables, metadata, and data editing. Attribute table: editing and querying data. Symbolization and categorization of geological features.	17
Unit 3	**GPS, Field Integration & Thematic Mapping** Global Positioning System (GPS): working principle, coordinate capture. Use of GPS in field data collection (Collecting GPS data using device/mobile app). Importing field data into GIS. Converting GPS data to shapefiles. Thematic map creation: lithology, land use, mineral occurrence, faults, etc. Map composition: layout design with north arrow, scale bar, and legend.	16
Unit 4	**Terrain Analysis, GIS Tools & Geological Applications** DEMs and terrain data: types, sources, geological significance. Loading DEM in GIS. Concept of slope, aspect, and hillshade maps. Generating Slope, aspect, hillshade, and drainage. Buffer, intersect, and overlay tools in GIS and their usage in geological queries. Applications of GIS in geological hazard mapping, site selection, and resource mapping. Mini-project: creation of a geology-related GIS map from multiple data sources.	17
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Credit Hours		**90**

Text Books suggested:

1) De Mars, M. N., 1999: Fundamentals of Geographic Information Systems, John Wiley & Sons Inc., New York.
2) Gopi, S., 2005: Global Positioning System Principles and Applications, Ta McGraw Hill, New Delhi.

Reference Books:

1) Curtis, H., 2000: The GPS Accuracy Improvement Initiative, GPS World, June, 2000.
2) Gonzalez, R. C., Woods, R. E., 2000: Digital Image Processing, Fifth Indian Reprint, Addison Wesley Longman, Delhi.

3) Miller, V. C.,1961: Photogeology; McGraw-Hill, New York.

Detailed Syllabus Of Semester 4

Type of Course: Major	Principles of Stratigraphy			Course Code:
	Course Level: 200	Credit: 3	L-T-P-C: 3-1-0-3	GEOL162M401
		Scheme of Evaluation: Theory		

Course Objectives: This course provides an introduction to the fundamental principles and concepts of stratigraphy. Students will learn about the methods and techniques used to study and interpret the layers of rocks that make up Earth's crust, including the principles of relative and absolute dating, correlation, and stratigraphic nomenclature. The course will also cover the major events and processes that have shaped Earth's geologic history, as recorded in the rock record.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Students will be able to recall and recognize the basic concepts, principles, and terminology of stratigraphy.	**BT 1**
CO 2	Students will be able to explain the fundamental processes and phenomena that shape the rock record, and the principles and methods used to study and interpret stratigraphic data.	**BT 2**
CO 3	Students will be able to apply stratigraphic principles and techniques to analyse and interpret geologic data and to reconstruct the geologic history of a region.	**BT 3**
CO 4	Students will be able to analyse the spatial and temporal relationships between rock units, and to evaluate the relative ages and depositional environments of these units.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	The scope and objectives of stratigraphy The major events and processes that have shaped Earth's geologic history The evolution of life on Earth and its relation to geologic events. The geologic time scale and its subdivisions.	17
Unit 2	Concepts of Lithostratigraphy, Chrono-stratigraphy and Bio-stratigraphy. The principles and guidelines for stratigraphic nomenclature (ICS code of nomenclature). Introductory concepts of sequence stratigraphy, chemo- stratigraphy and magneto-stratigraphy. Global Mass extinction events.	16
Unit 3	Concepts and methods of stratigraphic correlation. The use of biostratigraphy and chemo-stratigraphy in stratigraphic correlation. The Quaternary Period and its divisions, Neogene-Quaternary and Pleistocene- Holocene boundary, the Meghalayan Age.	17
Unit 4	Quaternary stratigraphy- principles and application in Quaternary sequences (Indian examples), soil profile and palaeosol, Quaternary records from marine and continental settings, event stratigraphy. Idea of Quaternary climate changes (glaciation and sea level changes).	16
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Notional Credit Hours		**90**

Text Books suggested:

1) Stratigraphic Principles and Practices – J. M. Weller; Universal Book Stall, Delhi.
2) Principles of Sedimentology and Stratigraphy, by Sam Boggs, Jr., 4th Edition, Pearson Prentice Hall, 2006.

Reference Books:

1) Stratigraphy: Principles and Methods by Stanley, Steven M.
2) Stratigraphy: A Modern Synthesis by Sloss, L. L.
3) The Geologic Time Scale 2020 by Gradstein, Felix M.
4) Sedimentary Geology: An Introduction to Sedimentary Rocks and Stratigraphy by Prothero, Donald R.
5) Basic Concepts in Sedimentology and Stratigraphy by Nichols, Gary.

Type of Course: Major	Palaeontology			Course Code:
	Course Level: 200	Credit: 3	L-T-P-C: 3-1-0-3	GEOL162M402
		Scheme of Evaluation: Theory		

Course Objectives: Palaeontology is the study of ancient life, focusing on the evolution, diversity, and extinction of organisms over geological time. This course will cover the history and methods of palaeontological research, the principles of evolutionary biology, and the study of fossils as evidence of past life. Topics covered will include the origin and evolution of life, major extinction events, the use of fossils in stratigraphy, and the interpretation of the ecological and biogeographic contexts of ancient ecosystems.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Students will be able to recall and recognize the key concepts, principles, and facts related to the study of palaeontology	**BT 1**
CO 2	Students will be able to explain the principles of evolutionary biology and the methods used in palaeontological research.	**BT 2**
CO 3	Students will be able to apply palaeontological principles to identify, describe, and interpret the significance of fossils in the context of past life and environments.	**BT 3**
CO 4	Students will be able to analyse the morphology, diversity, and distribution of fossil organisms and their significance in the evolutionary history of life.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	Nature and importance of fossil record; Fossilization processes and modes of preservation. Types of fossils (body fossils, trace fossils, leaked fossils, etc.). Importance of Index fossils. Theory of organic evolution as interpreted from fossil record. Speciation, Taxonomic hierarchy. Introduction to Palae-botany and Ichnology.	16
Unit 2	Morphology and stratigraphic significance of the following important invertebrate groups: a. Brachiopoda b. Lamellibranchia (Bivalvia) c. Cephalopoda d. Gastropoda e. Echinoidia f. Trilobites.	17
Unit 3	Origin of vertebrates and major steps in vertebrate evolution. Brief introduction to vertebrate palaeontology (Hominidae, Equidae, Proboscidae). Mesozoic reptiles with special reference to origin, diversity and extinction of dinosaurs.	17
Unit 4	Microfossils and its application. Introduction to palynology. Detailed account of Gondwana Flora in India in the context of Palaeoclimate and Palaeoecology. Role of fossils in sequence stratigraphy, hydrocarbon exploration and palaeoclimatic studies. Fossils and biogeographic provinces, dispersals and barriers.	16
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Notional Credit Hours		**90**

Text Books suggested:

1) An Introduction to Palaeontology – Amal Dasgupta, The World Press Private Limited.
2) Palaeontology: (Palaeobiology) Evolution and Animal Distribution – P.C. Jain, M.S. Anantharaman, Vishal Publishing.

Reference Books:

1) Introduction to Paleobiology and the Fossil Record – Michael J. Benton, David A. T. Harper, and Robert L. Carroll, 2nd edition, 2013 by Wiley-Blackwell.
2) Principles of Paleontology - D. M. Raup & S. M. Stanley. W. H. Freeman (1971).
3) Fossils in Earth Sciences – Anis Ray.

Type of Course: Major	Earth Science in Ancient India	Course Code:
	Course Level: 200 Credit: 4 L-T-P-C: 3-1-0-4 Scheme of Evaluation: Theory	GEOL162M403

Course Objectives: This course aims to explore the rich heritage of earth science in ancient India, delving into the cosmological, geological, and environmental knowledge embedded in Vedic and other ancient Indian texts. By examining the ancient Indian perspectives on the Earth, the solar system, and natural phenomena, students will develop a deeper understanding of the interconnectedness between ancient wisdom and modern scientific principles.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Students will acquire knowledge of Vedic cosmology and ancient Indian astronomical theories, including the concepts of Lokas, Nakshatras, and the contributions of ancient Indian astronomers to celestial mechanics.	**BT 1**
CO 2	Students will comprehend the geological significance of minerals, metals, and landforms described in ancient Indian texts, as well as the hydrological and geomorphic features of the Sapta Sindhu region.	**BT 2**
CO 3	Students will apply their understanding of ancient Indian tectonic theories and geomantic concepts to interpret the formation of mountains, earthquakes, and architectural principles in Vastu Shastra.	**BT 3**
CO 4	Through the analysis of ancient water management systems and environmental ethics embedded in ancient Indian texts, students will critically evaluate the sustainability and conservation practices advocated by ancient Indian civilization.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	**The Earth and the Solar System**: Vedic texts for references to cosmology, celestial bodies, and their significance. Nakshatras (lunar mansions) and their connection to timekeeping. Works of ancient Indian astronomers such as Aryabhata, Varahamihira, and Brahmagupta: their contributions to understanding planetary motion, eclipses, and celestial coordinates. Siddhantas (astronomical treatises) and their mathematical models.	20
Unit 2	**Earth Materials, Surface Features, and Processes**: Examine ancient Indian texts for references to minerals, metals, and their uses. Significance of minerals like gold, copper, and iron in ancient trade and economy. Ancient mining techniques and sites of India. Descriptions of landforms in ancient texts. Investigate the Brahmaputra Valley and its geological features. Role of rivers (e.g., Sarasvati, Ganga, Brahmaputra, etc) in shaping the landscape.	20
Unit 3	**Interior of the Earth, Deformation, and Tectonics**: Investigate ancient Indian theories on the formation of mountains and earthquakes. Concept of Meru and its geological symbolism. Study ancient texts for accounts of earthquakes. Introduction to Geo-archaeology.	25
Unit 4	**Natural resource management and sustainable livelihood**: Ancient water management systems (e.g., stepwells, irrigation tanks, aqueducts). Importance of water conservation and sustainable practices. References to rainfall patterns and flood control. Description of Sea-level change in ancient texts. Ancient urban planning, sustainable housing, etc. Ancient metallurgy and mineral beneficiation.	25
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		30
Total Notional Credit Hours		**90**

Reference Books:

1) "The Lost River: On The Trail of the Sarasvati" by Michel Danino
2) "Vedic Cosmology: Mysteries of the Sacred Universe" by Richard L. Thompson
3) "Astronomy in India: A Historical Perspective" by Mani Bhaumik
4) "Ancient India as Described by Megasthenes and Arrian: Being a Translation of the Fragments of the Indika of Megasthenes" by J. W. McCrindle
5) “Ancient mining techniques and sites of India” By A. K. Grover, Retd. DDG GSI.
6) A Textbook of Rasashastra’ by Dr. Anuroopa H.K.

Type of Course: Major	Stratigraphy and Palaeontology Practical	Course Code:
	Course Level: 200 Credit: 2 L-T-P-C: 0-1-4-2 Scheme of Evaluation: Practical	GEOL162M414

Course Objectives: The course aims to develop hands-on skills in interpreting geological and paleontological data through construction of stratigraphic sections, diagrams, and columns. It emphasizes fossil identification, evolutionary trends, and the reconstruction of past environments using real and simulated datasets. Students will gain the ability to integrate stratigraphic and paleontological evidence for understanding Earth's history and correlating geological formations.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recognize and name common fossils, stratigraphic terms, and basic sedimentary structures.	**BT 1**
CO 2	Explain the relationship between fossil assemblages and stratigraphic horizons.	**BT 2**
CO 3	Construct stratigraphic columns, fence diagrams, and geologic sections from raw data.	**BT 3**
CO 4	Interpret evolutionary trends and paleoenvironments using fossil records and stratigraphic data.	**BT 4**

Modules	Topics and Course Content	Hours
I	**Construction and Interpretation of Geological Cross Sections:** Using geological maps and subsurface data infer stratigraphic relationships and structures. **Reconstruction of Proterozoic Supercontinents through Time:** Visualisation of paleogeographic maps; interpreting tectonic and stratigraphic evolution. **Preparation of Fence Diagrams from Stratigraphic Logs:** Correlation of lithostratigraphic units from multiple boreholes or sections. **Development of Stratigraphic Columns and Paleoenvironments:** Stratigraphic column construction from given lithologs and inference of depositional settings. **Documentation of stratigraphic details in the field.**	30
II	**Megafossil Identification from Hand Specimens:** Morphological and taxonomic identification of common invertebrate and vertebrate fossils. **Determination of Evolutionary Trends in Fossil Series:** Analyzing morphological changes in an evolutionary lineage (e.g., *trilobites*, *ammonites*, *horses*, etc.). **Identification of Fossil Assemblages and Their Stratigraphic Significance:** Assigning fossil groups to biozones and interpreting their age and depositional settings. **Microscopic Identification of Microfossils Using Slides and Microscope:** Identification of foraminifera, radiolaria, ostracods, etc., with classification and environmental interpretation. **Palynological Exercises: Study of Spores and Pollen Grains:** Extraction techniques (demo or data-based), identification, and use in biostratigraphy and palaeoenvironmental analysis. **Identification of fossils and fossil assemblages in field.**	30
Total Credit Hours		**60**

Text Books suggested:

1) Principles of Sedimentology and Stratigraphy, by Sam Boggs, Jr., 4th Edition, Pearson Prentice Hall, 2006.
2) Palaeontology: (Palaeobiology) Evolution and Animal Distribution – P.C. Jain, M.S. Anantharaman, Vishal Publishing.

Reference Books:

1) Introduction to Paleobiology and the Fossil Record – Michael J. Benton, David A. T. Harper, and Robert L. Carroll, 2nd edition, 2013 by Wiley-Blackwell.
2) Sedimentary Geology: An Introduction to Sedimentary Rocks and Stratigraphy by Prothero, Donald R.

Type of Course: Minor	Petrology Course Level: 200 Credit: 3 L-T-P-C: 3-1-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162N401

Course Objectives: To provide students with a comprehensive understanding of the formation, classification, and characteristics of igneous, metamorphic, and sedimentary rocks, enabling them to interpret petrological processes and their implications for Earth's geological history and tectonic evolution.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Students will be able to demonstrate a thorough understanding of the classification, textures, and structures of igneous, metamorphic, and sedimentary rocks, as well as the geological processes responsible for their formation.	**BT 1**
CO 2	Students will be able to interpret and explain the physical and chemical properties of magmas, the factors controlling metamorphism, and the processes involved in weathering and sedimentary flux, demonstrating a deeper comprehension of petrological concepts.	**BT 2**
CO 3	Students will be able to apply their knowledge of petrology to identify and interpret various types of igneous, metamorphic, and sedimentary rocks in hand specimen and thin section, and to relate these observations to geological environments and tectonic settings.	**BT 3**
CO 4	Students will develop the ability to analyse petrological data, including mineral assemblages, textures, and structures, to infer past geological conditions, such as temperature, pressure, and deformation regimes, and to evaluate the processes involved in rock formation and modification.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	**Igneous Petrology:** Introduction: Heat flow, geothermal gradient, Physical and chemical properties of magmas. Classification of igneous rocks. Textures and structures of igneous rocks. Mode of occurrence of Igneous rocks. Crystallisation of Magma, Reaction Principle, Magmatic differentiation. Plate tectonics and igneous rock formation.	9
Unit 2	**Metamorphic Petrology:** Definition of metamorphism and Metasomatism. Factors controlling metamorphism. Types of metamorphism - contact metamorphism, regional metamorphism, fault zone metamorphism, impact metamorphism. Index minerals, Metamorphic zones and isograde. Concept of metamorphic facies and grade. Structure and textures of metamorphic rocks. and role of fluids in metamorphism.	9
Unit 3	**Sedimentology:** Weathering and sedimentary flux: Physical and chemical weathering. Sedimentary texture: size, shape, roundness, sphericity, fabric, packing. Concepts of diagenesis, Stages of diagenesis, Compaction and cementation. Textural classification of sediments and sedimentary rocks.	9
Unit 4	Sediment dynamics: Nature of fluid flow – Laminar vs. turbulent flow, concept of flow regime and sediment transport. Sedimentary structures – bedforms and internal stratification. Concept of sedimentary environment and facies.	9
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Notional Credit Hours		**60**

Text Books suggested:

1) Principles of Petrology – G. W. Tyrrell. (1926). Springer
2) Principles of igneous and metamorphic petrology – J. D. Winter (2014). Pearson
3) Introduction to Sedimentology – S. M. Sengupta, (2018), CBS.

Reference Books:

1) Principles of igneous and metamorphic petrology – A. Philpotts & J. Ague. (2009). Cambridge University Press.
2) Sedimentary Rocks – F. J. Pettijohn

Type of Course: Minor	Structural Geology Course Level: 200 Credit: 3 L-T-P-C: 3-1-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162N402

Course Objectives: Structural geology is a sub-discipline of geology that deals with the study of deformation and deformation-related structures of rocks at various scales. This course aims to provide a fundamental understanding of structural geology, including the analysis of structural data and the interpretation of deformation processes that occur in the Earth's crust.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Students will be able to recall the basic concepts, terminology, and principles of structural geology.	**BT 1**
CO 2	Students will be able to comprehend the various types of rock deformation and deformation related structures, including folds, faults, and joints.	**BT 2**
CO 3	Students will be able to apply their knowledge of structural geology to analyze and interpret geological maps and cross-sections.	**BT 3**
CO 4	Students will be able to analyse structural data, including the measurement and plotting of various structural elements, such as strike, dip, and plunge.	**BT 4**

Modules	Topics and Course Content	Hours
Unit 1	Diastrophic and non-diastrophic structures. Structural elements: planar and linear structures. Concept of strike and dip, trend and plunge, rake/pitch. Outcrop patterns of different structures.	9
Unit 2	**Concept of rock deformation** Stress – normal and shear stress, stress at a point. Stress ellipsoid and principal stress axes. Mohr's stress circle and various stress types. Strain in rocks, types of strain. Principal strain axes and Strain ellipses. Flinn's diagram.	9
Unit 3	Concept of brittle and ductile deformation. Fold morphology; Geometric and genetic classification of folds. Introduction to the mechanics of folding: Buckling, Bending, Flexural slip and flow folding. Description and origin of foliations and lineations.	9
Unit 4	Geometry of pinch and swell and boudin structure. Basic idea of shear zone, faults and joints. Geometric and genetic classification of fractures and faults. Geologic/geomorphic criteria for recognition of faults.	9
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Notional Credit Hours		**60**

Text Books suggested:

1) Structural Geology - Robert J. Twiss & Eldridge M. Moores, (2nd edition, 2007), W. H. Freeman & Co Ltd.
2) Structural Geology - M. P. Billings, 4th edition, Prentice-Hall.

Reference Books:

1) Foundations of Structural Geology - Park, R. G. (2005), Routledge.
2) Structural Geology – Fundamentals and Modern Developments - S. K. Ghosh, (2013), Elsevier Science.
3) Structural Geology of Rocks and Region - G. R. Davis, (1984), John Wiley.
4) Structural Geology – Haakon Fossen, (2010), Cambridge University Press, New York.

Type of Course: AEC	Business Communication: Concepts and Skills Credit: 2 L-T-P-C: 2-1-0-2 Scheme of Evaluation: Theory	Course Code: CEN982A401

Course Objectives: This course is designed to enhance employability and maximize the students' potential by introducing them to the principles that determine personal and professional success, thereby helping them acquire the skills needed to apply these principles in their lives and careers.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Demonstrate understanding the importance of verbal and non-verbal skills while delivering an effective presentation.	**BT 2**
CO 2	Develop professional documents to meet the objectives of the workplace	**BT 3**
CO 3	Identify different life skills and internet competencies required in personal and professional life.	**BT 3**

Modules	Topics and Course Content	Hours
Unit 1	**Presentation Skills** Importance of presentation skills, Essential characteristics of a good presentation, Stages of a presentation, Visual aids in presentation, Effective delivery of a presentation.	5
Unit 2	**Business Writing** Report writing: Importance of reports, Types of reports, Format of reports, Structure of formal reports Proposal writing: Importance of proposal, Types of proposal, structure of formal proposals Technical articles: Types and structure	5
Unit 3	**Preparing for jobs** Employability and Unemployability, Bridging the Industry-Academia Gap Knowing the four-step employment process, writing resumes, Guidelines for a good resume, Writing cover letters Interviews: Types of interviews, what does a job interview assess, strategies of success at interviews, participating in group discussions.	5
Unit 4	**Digital Literacy and Life Skills** Digital literacy: Digital skills for the '21st century', College students and technology, information management using Webspace, Dropbox, directory, and folder renaming conventions. Social Media Technology and Safety, Web 2.0. **Life Skills:** Overview of Life Skills: Meaning and significance of life skills, Life skills identified by WHO: self-awareness, Empathy, Critical thinking, Creative thinking, Decision making, problem- solving, Effective communication, interpersonal relationship, coping with stress, coping with emotion. Application of life skills: opening and operating bank accounts, applying for PAN, Passport, online bill payments, ticket booking, gas booking	5
EL	- Movie/ Documentary screening - Field visits - Peer teaching - Seminars - Library visits	10
Total Notional Credit Hours		**30**

Text Books:

1) Business Communication by Shalini Verma
2) Technical Communication by Meenakshi Raman and Sangeeta Sharma

Detailed Syllabus Of Semester 5

Type of Course: Major	Structural Geology Course Level: 300 Credit: 3 L-T-P-C: 3-0-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162M501

Course Objectives: The course aims to provide students with a fundamental understanding of rock deformation processes, structural analysis techniques, and their applications in tectonics, resource exploration, and geohazard assessment.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall fundamental concepts of structural geology, including types of rock deformation, stress-strain relationships, and structural elements.	**BT 1**
CO 2	Explain the fundamental concepts of rock deformation, stress, and strain, and their role in shaping Earth's crust.	**BT 2**
CO 3	Identify and classify various structural features such as folds, faults, joints, foliations, and lineations based on their geometry, genesis, and mechanics.	**BT 3**
CO 4	Analyse the mechanics of brittle and ductile deformation, including fault kinematics, shear zones, and superposed deformation.	**BT 4**
CO 5	Interpret structural data using stereographic projections, rose diagrams, and cross-section construction for geological mapping and resource exploration.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Fundamentals of Structural Geology** Introduction to Structural Geology: Importance in geosciences and resource exploration. Structural Elements: Planar and linear features, strike and dip, trend and plunge, rake/pitch. Rock Deformation Concepts: Brittle vs. ductile deformation, types of deformation (elastic, plastic, and rupture). Stress and Strain: • Types of stress (normal, shear, hydrostatic, differential). • Stress ellipsoid, principal stress axes, Mohr's stress circle. • Strain ellipsoid, principal strain axes, types of strain, Flinn's diagram.	11
Unit 2	**Brittle Structures** Fractures and Joints: Classification, formation mechanisms, tectonic significance. Faults: • Classification (dip-slip, strike-slip, oblique-slip, listric, growth faults). • Mechanics of faulting and fault plane characteristics. • Fault zone structures: gouge, breccia, mylonites. • Criteria for fault recognition in the field.	11
Unit 3	**Ductile Deformation & Folding** Folds: • Geometry and classification (geometric and genetic). • Fold mechanisms: buckling, bending, flexural slip, flexural flow, passive folding. • Interference patterns of folds. Foliation & Lineation: • Types, formation processes, relation to stress fields. • Shear fabrics (S-C structures, mineral lineations). Boudinage: Types, formation processes, tectonic significance.	11
Unit 4	**Structural Analysis and Applications** Geometric Analysis of Structures: • Principles of structural analysis and representation in maps. • Outcrop patterns of folded, faulted, and unconformable sequences. • Importance of cross-section construction in structural interpretation. Tectonic Controls on Rock Structures:	12

	• Relationship between structural geology and regional tectonics. • Structural control on mineralisation, hydrocarbon reservoirs, and groundwater flow. Introduction to Digital Structural Mapping: • Use of GIS and remote sensing for structural interpretation. • Role of digital elevation models (DEMs) in structural geology.	
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		45
Total Credit Hours		**90**

Text Books suggested:

1) Fossen, H. (2016). Structural Geology. Cambridge University Press.
2) Twiss, R. J., & Moores, E. M. (2007). Structural Geology. W. H. Freeman.

Reference Books:

1) Davis, G. H., Reynolds, S. J., & Kluth, C. (2011). Structural Geology of Rocks and Regions. Wiley.
2) Van der Pluijm, B. A., & Marshak, S. (2004). Earth Structure: An Introduction to Structural Geology and Tectonics. W. W. Norton.
3) Ramsay, J. G. (1967). Folding and Fracturing of Rocks. McGraw-Hill.
4) Lisle, R. J., Brabham, P. J., & Barnes, J. W. (2011). Basic Geological Mapping. Wiley-Blackwell.

Type of Course: Major	Structural Geology Practical - I Course Level: 300 Credit: 1 L-T-P-C: 0-0-2-1 Scheme of Evaluation: Practical	Course Code: GEOL162M512

Course Objectives: The course aims to provide students with a fundamental understanding of rock deformation processes, structural analysis techniques, and their applications in tectonics, resource exploration, and geohazard assessment.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall fundamental concepts of structural geology, including types of rock deformation, stress-strain relationships, and structural elements.	**BT 1**
CO 2	Explain the fundamental concepts of rock deformation, stress, and strain, and their role in shaping Earth's crust.	**BT 2**
CO 3	Identify and classify various structural features such as folds, faults, joints, foliations, and lineations based on their geometry, genesis, and mechanics.	**BT 3**
CO 4	Analyse the mechanics of brittle and ductile deformation, including fault kinematics, shear zones, and superposed deformation.	**BT 4**
CO 5	Interpret structural data using stereographic projections, rose diagrams, and cross-section construction for geological mapping and resource exploration.	**BT 5**

Modules	Topics and Course Content	Hours
I	**Plotting and Analysis of Planar and Linear Structures** – Using stereographic projections and structural data. **Mohr's Circle Construction** – For 2D stress conditions; graphical solution of stress problems. **Strain Analysis** – Using strain ellipse, principal strain axes, and plotting on Flinn diagram. **Identification and Interpretation of Folds** – Using block diagrams, models, and geological maps. **Preparation and Interpretation of Geological Maps** – Construction of cross-sections, identification of faults, folds, and unconformities. **Thickness Calculations of strata** – True and apparent thickness from exposures and maps **Recognition and Classification of Faults** – With the help of photographs, maps, and field examples. **Three-Point Borehole Problems** – Dip and strike calculations from the given set of maps. **Field visit for a period of 1 week from 1st to 7th November** – Identification and documentation of folds, faults, and joints in local outcrops.	30
Total Credit Hours		**30**

Text Books suggested:

1) Fossen, H. (2016). Structural Geology. Cambridge University Press.
2) Twiss, R. J., & Moores, E. M. (2007). Structural Geology. W. H. Freeman.

Reference Books:

1) Davis, G. H., Reynolds, S. J., & Kluth, C. (2011). Structural Geology of Rocks and Regions. Wiley.
2) Ramsay, J. G. (1967). Folding and Fracturing of Rocks. McGraw-Hill.
3) Lisle, R. J., Brabham, P. J., & Barnes, J. W. (2011). Basic Geological Mapping. Wiley-Blackwell.

Type of Course: Major	Indian Stratigraphy Course Level: 300 Credit: 4 L -T-P-C: 3-1-0-4 Scheme of Evaluation: Theory	Course Code: GEOL162M503

Course Objectives: To develop a fundamental understanding of the stratigraphic framework of India, including its cratonic provinces, sedimentary basins, and volcanic provinces, while also exploring the major stratigraphic boundaries and their significance in paleoenvironmental and climate change studies.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall the physiographic divisions of India and their relation to major geological provinces.	**BT 1**
CO 2	Explain the geological and tectonic history of India, including major cratons, mobile belts, and sedimentary basins.	**BT 2**
CO 3	Apply fundamental stratigraphic principles to analyse and correlate Proterozoic, Phanerozoic, and Quaternary successions in India.	**BT 3**
CO 4	Distinguish between various volcanic provinces and stratigraphic boundaries in India and assess their economic and geodynamic significance.	**BT 4**
CO 5	Evaluate the role of Quaternary stratigraphy in reconstructing past climatic conditions and predicting future climate change trends.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	Brief introduction to the physiographic subdivisions of India. Introduction to Indian Shields- Cratons, Mobile belts and Shear zones Geological and tectonic history of India.	14
Unit 2	Introduction to Proterozoic basins of India: Vindhyans and Cuddapah SuperGroup Brief geology of Dharwar, Bastar, Singhbhum, Aravalli, Bundelkhand, Eastern Ghat Mobile Belt and Shillong Plateau.	18
Unit 3	**Palaeozoic stratigraphy** of Kashmir and its correlatives from Spiti and Zanskar. **Gondwana Stratigraphy.** **Mesozoic stratigraphy of India:** Triassic of Spiti, Jurrasic of Kutch and Cretaceous successions of Cauvery basin. **Cenozoic stratigraphy of India:** Siwalik successions, Assam-Arakan basin **Volcanic provinces of India:** Deccan Trap, Rajmahal Trap, Sylhet Trap, Abor Volcanics.	20
Unit 4	**Important Stratigraphic boundaries in India:** a. Precambrian-Cambrian boundary b. Permian-Triassic boundary c. Cretaceous-Paleogene boundary **Introduction to Quaternary Stratigraphy:** Glacial and interglacial deposits in Himalayas, Fluvial and Aeolian sediments of Thar Desert, and Indo-Gangetic-Brahmaputra Plain. Meghalayan Age and its significance. Significance of Quaternary stratigraphy in climate change studies.	20
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		48
Total Credit Hours		**120**

Text Books suggested:

1) Valdiya, K. S. (2010). The Making of India: Geodynamic Evolution. Springer.
2) Krishnan, M. S. (2009). Geology of India and Burma (6th Edition). CBS Publishers & Distributors.

Reference Books:

1) Indian Stratigraphy by Srikant Das, Birbal Sahni Institute of Paleobotany (2018)
2) Indian Geology: An Introduction by D. N. Wadia, Tata McGraw-Hill Education (2007)
3) Geology of India by V. P. Dimri, Springer (2020)
4) Geology of India (Vol. 1 & 2) – M. Ramakrishnan & R. Vaidyanadhan, Geological Society of India, Bangalore (2008).

Type of Course: Major	Hydrogeology			Course Code: GEOL162M504
	Course Level: 300	Credit: 3	L -T-P-C: 3-0-0-3	
		Scheme of Evaluation: Theory		

Course Objectives: To provide students with a comprehensive understanding of groundwater occurrence, movement, exploration, chemistry, and management, enabling them to apply hydrogeological concepts for sustainable groundwater resource utilization.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall fundamental concepts of hydrogeology, including the hydrologic cycle, types of aquifers, and groundwater movement.	**BT 1**
CO 2	Explain the fundamental concepts of the hydrologic cycle, groundwater occurrence, and aquifer systems.	**BT 2**
CO 3	Apply Darcy's law and well hydraulics principles to analyse groundwater flow and aquifer properties.	**BT 3**
CO 4	Analyse different groundwater exploration techniques and their applications in hydrogeological studies.	**BT 4**
CO 5	Evaluate groundwater quality by interpreting hydrochemical parameters and contamination risks.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Introduction** Scope of hydrogeology and its societal relevance. Hydrologic cycle: precipitation, evapo-transpiration, run-off, infiltration and subsurface movement of water. Rock properties affecting groundwater, Vertical distribution of subsurface water. Types of aquifers, aquifer parameters. Groundwater provinces of India.	16
Unit 2	**Groundwater flow and Well hydraulics** Darcy's law and its validity; Intrinsic permeability and hydraulic conductivity. Groundwater flow rates and flow direction; Laminar and turbulent groundwater flow. Basic Concepts of Well hydraulics (drawdown; specific capacity etc).	17
Unit 3	**Groundwater exploration and Hydrochemistry** Surface-based groundwater exploration methods. Introduction to subsurface borehole logging methods. Physical and chemical properties of water and water quality.	16
Unit 4	**Groundwater management** Groundwater recharge and conservation strategies. Surface and subsurface water interaction. Groundwater pollution: sources, effects, and remediation. Groundwater management in urban areas Climate change and its impact on groundwater resources. Groundwater occurrences in crystalline rocks.	17
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Credit Hours		**90**

Text Books suggested:

1) Todd, D.K. & Mays, L.W. (2005). Groundwater Hydrology (3rd Edition). Wiley.
2) Raghunath, H.M. (2007). Groundwater (3rd Edition). New Age International Publishers.

Reference Books:

1) Fetter, C.W. (2001). Applied Hydrogeology (4th Edition). Prentice Hall.
2) Karanth, K.R. (1987). Groundwater Assessment, Development and Management. Tata McGraw-Hill.
3) Domenico, P.A. & Schwartz, F.W. (1998). Physical and Chemical Hydrogeology (2nd Edition). Wiley.

Type of Course: Major	Hydrogeology Practical Course Level: 300 Credit: 1 L -T-P-C: 0-0-2-1 Scheme of Evaluation: Practical	Course Code: GEOL162M515

Course Objectives: To provide students with a comprehensive understanding of groundwater occurrence, movement, exploration, chemistry, and management, enabling them to apply hydrogeological concepts for sustainable groundwater resource utilization.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall fundamental concepts of hydrogeology, including the hydrologic cycle, types of aquifers, and groundwater movement.	**BT 1**
CO 2	Explain the fundamental concepts of the hydrologic cycle, groundwater occurrence, and aquifer systems.	**BT 2**
CO 3	Apply Darcy's law and well hydraulics principles to analyse groundwater flow and aquifer properties.	**BT 3**
CO 4	Analyse different groundwater exploration techniques and their applications in hydrogeological studies.	**BT 4**
CO 5	Evaluate groundwater quality by interpreting hydrochemical parameters and contamination risks.	**BT 5**

Modules	Topics and Course Content	Hours
I	**Preparation and Interpretation of Water Table Contour Maps and depth to water level maps.** – Construction of potentiometric surface and inference of groundwater flow direction. **Calculation of Aquifer Parameters** – Permeability, Transmissivity, storativity, specific yield, groundwater flow, and well hydraulics using given data. **Determination of Specific Capacity and Drawdown** – From well test data (pumping test/case studies). **Classification of Aquifers** – Interpretation of geological cross-sections to identify unconfined, confined, and perched aquifers. **Estimation of Water Quality Parameters** – TDS, pH, hardness, chloride, and nitrate using test kits or secondary data.	30
Total Credit Hours		**30**

Text Books suggested:

1) Todd, D.K. & Mays, L.W. (2005). Groundwater Hydrology (3rd Edition). Wiley.
2) Raghunath, H.M. (2007). Groundwater (3rd Edition). New Age International Publishers.

Reference Books:

1) Fetter, C.W. (2001). Applied Hydrogeology (4th Edition). Prentice Hall.
2) Karanth, K.R. (1987). Groundwater Assessment, Development and Management. Tata McGraw-Hill.
3) Domenico, P.A. & Schwartz, F.W. (1998). Physical and Chemical Hydrogeology (2nd Edition). Wiley.

Type of Course: Major	Summer Internship Credit: 4 Scheme of Evaluation: Project	Course Code: GEOL162M526

Course Objectives: To provide students with hands-on experience in geological fieldwork, laboratory techniques, and industry applications, bridging academic knowledge with real-world geological practices.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Explain the practical applications of geological concepts in real-world scenarios through hands-on fieldwork, laboratory analysis, or industrial training.	**BT 2**
CO 2	Demonstrate technical skills in geological mapping, exploration techniques, hydrogeological assessments, and geotechnical investigations in professional settings.	**BT 3**
CO 3	Develop professional communication skills by preparing structured reports, maintaining field diaries, and delivering presentations based on internship experiences.	**BT 3**
CO 4	Assess geological data, interpret field observations, and evaluate resource potential, environmental impacts, or structural stability in diverse geological terrains.	**BT 4**
CO 5	Critically examine the relevance of geological methods and tools used in industry and research, applying problem-solving skills to tackle real-world challenges.	**BT 5**

Internship Structure & Guidelines

1

Internship Duration & Timeline:

- **Total Duration:** 45 Days
- **Internship Period:** After completion of 4th semester examinations and before the commencement of the 5th semester.

2

Internship Venues:

- **Government Organisations:** GSI, AMD, CGWB, NHPC, ISRO, NGRI, IMD, etc.
- **Mining & Exploration Companies:** Coal India, MECL, HZL, Vedanta, OIL, ONGC, etc.
- **Geotechnical & Environmental Firms:** CBRI, WAPCOS, NEERI, and other consultancies.
- **Academic & Research Institutions:** Universities, IITs, CSIR labs, and NGOs working in earth sciences and disaster management.

3

Nature of Work:

- **Geological Mapping & Fieldwork:** Lithological mapping, core logging, structural analysis.
- **Mineral & Petroleum Exploration:** Ore reserve estimation, petrography, hydrocarbon exploration techniques.
- **Hydrogeology & Environmental Studies:** Water resource assessments, groundwater quality analysis, remote sensing applications.
- **Geotechnical & Engineering Geology:** Slope stability studies, site investigations, tunnelling, and construction material testing.
- **GIS & Remote Sensing:** Digital mapping, satellite image interpretation, DEM analysis.

4

Assessment & Evaluation (100 Marks)

Component	Weightage (%)
Internship Diary (Daily work record)	15%
Mid-Term Report (Progress update)	20%
Final Internship Report (Comprehensive documentation)	30%
Presentation & Viva-Voce (Evaluation by faculty panel)	35%

- **Internship Diary:** Daily record of observations, tasks, and learnings.
- **Mid-Term Report:** A brief progress report submitted midway through the internship.
- **Final Report:** Detailed documentation of the work carried out, including methodologies, results, and conclusions.
- **Presentation & Viva-Voce:** Oral presentation of findings before a faculty panel, followed by a Q&A session.

Type of Course: Minor	Fuel Geology Course Level: 200 Credit: 4 L -T-P-C: 3-0-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162N501

Course Objectives: To provide fundamental knowledge of the origin, occurrence, exploration, and economic significance of coal, petroleum, gas hydrates, and nuclear fuels, with a focus on their geological characteristics and global/Indian distribution.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall the fundamental concepts of coal, petroleum, gas hydrates, and nuclear fuels, including their origin, classification, and occurrence.	**BT 1**
CO 2	Understand the formation processes, geological settings, and economic significance of various fuel resources.	**BT 2**
CO 3	Apply knowledge of fuel geology in resource exploration and extraction techniques.	**BT 3**
CO 4	Analyse different types of hydrocarbon traps, reservoir characteristics, and the impact of plate tectonics on fuel distribution.	**BT 4**
CO 5	Evaluate the sustainability, environmental impact, and future prospects of conventional and unconventional fuel resources.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Coal Geology** Definition and origin of coal; Process of coal formation (peatification, coalification). Basic classification of coal (Rank, Type, and Grade). Fundamentals of Coal Petrology: Lithotypes, Microlithotypes, and Macerals. Proximate and Ultimate analysis of coal. Introduction to Coal Bed Methane (CBM) Concept of Green Energy and Net Zero Goals.	12
Unit 2	**Petroleum Geology** Chemical composition and physical properties of crude oil. Origin and formation of petroleum: Organic matter transformation, maturation of kerogen. Migration of petroleum: Reservoir rocks, Cap rocks and Traps. Primary and secondary migration mechanisms.	12
Unit 3	**Petroleum reservoirs, traps and Gas Hydrates:** • Definition and significance of hydrocarbon traps. • Classification of hydrocarbon traps – Structural, stratigraphic, and combination traps. **Plate tectonics and global distribution of hydrocarbon reserves.** **Concept of Gas Hydrates**	12
Unit 4	**Introductory idea about Nuclear Fuels** • Uranium and Thorium- Mineralogy, occurrence and distribution in India. **Exploration and Detection of Nuclear Fuels:** • Radioactivity detectors: Geiger-Müller counter, proportional and scintillation counters, and gamma-ray spectrometers.	12
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		42
Total Credit Hours		**90**

Text Books suggested:

1) Chandra, D. Singh, R.M. & Singh, M.P. – Textbook of Coal (Indian Context) (Tata McGraw Hill)
2) Levorsen, A.I. – Geology of Petroleum (CBS Publishers)

Reference Books:

1) Tissot, B.P. & Welte, D.H. – Petroleum Formation and Occurrence (Springer)
2) Selley, R.C. – Elements of Petroleum Geology (Academic Press)
3) Durrance, E.M. – Radioactivity in Geology: Principles and Applications (Ellis Horwood Ltd.)
4) Dadhich, R.K. & Sharma, N.L. – Nuclear Fuel Cycle (Narosa Publishing House)
5) Bachu, S. – Gas Hydrates: Energy Resource and Environmental Challenges (Elsevier)

Detailed Syllabus Of Semester 6

Type of Course: Major	Geostatistics Course Level: 300 Credit: 4 L -T-P-C: 3-1-0-4 Scheme of Evaluation: Theory	Course Code: GEOL162M601

Course Objectives: The course aims to provide students with a fundamental understanding of geostatistical methods for spatial data analysis, focusing on their applications in geology, mining, hydrogeology, and environmental sciences. It introduces key concepts such as spatial variability, variogram analysis, interpolation techniques, and decision-making tools to enable students to analyse and model geological data effectively.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall fundamental concepts of statistics and geostatistics, including spatial data types and variability.	**BT 1**
CO 2	Understand the principles of spatial data analysis, correlation, regression, and hypothesis testing in geostatistics.	**BT 2**
CO 3	Apply geostatistical techniques such as variogram analysis, spatial autocorrelation, and interpolation methods to geological datasets.	**BT 3**
CO 4	Analyse geostatistical models for resource estimation, hydrogeology, and environmental applications using real-world case studies.	**BT 4**
CO 5	Evaluate and compare different geostatistical methods, including Kriging, inverse distance weighting, and machine learning-based approaches, for decision-making in geological studies.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Fundamentals of Geostatistics** Definition, scope, and significance of geostatistics. Applications in geology, mining, hydrogeology, and petroleum exploration. Basic concepts: spatial variability, random variables, stationarity, and spatial dependence. Types of spatial data: point data, areal data, volumetric data. Data exploration and visualization techniques (e.g., histograms, scatter plots, box plots).	16
Unit 2	**Statistical Methods in Geostatistics** Descriptive statistics: mean, median, mode, standard deviation, skewness, kurtosis, variance, and covariance. Correlation and regression analysis in spatial datasets. Probability distributions and their geological applications. Students' t-test, and Chi-square test.	16
Unit 3	**Spatial Interpolation and Variogram Analysis** Concept of variogram and semivariogram. Calculation of experimental variograms from spatial data. Interpretation of variograms: nugget effect, sill, and range. Structural analysis of spatial datasets: anisotropy and spatial continuity. Introduction to spatial interpolation techniques: inverse distance weighting, spline interpolation Kriging techniques: ordinary kriging, universal kriging, and indicator kriging.	18
Unit 4	**Advanced Applications and Decision-Making in Geostatistics** Geostatistical simulation techniques in reservoir modelling, mineral resource exploration, and mining modelling. Application of machine learning and AI in geostatistical analysis. Multi-criteria decision analysis using geostatistical methods. Introduction to modern geostatistical software tools (e.g., R, Python, QGIS, ArcGIS). Case studies on geostatistical applications in natural resource exploration and environmental management.	16
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Notional Credit Hours		**90**

Text Books suggested:

1) Davis, J.C. (2002). Statistics and Data Analysis in Geology. Wiley.
2) Chilès, J.P., & Delfiner, P. (2012). Geostatistics: Modeling Spatial Uncertainty. Wiley.

Reference Books:

1) Goovaerts, P. (1997). Geostatistics for Natural Resources Evaluation. Oxford University Press.
2) Kitanidis, P.K. (1997). Introduction to Geostatistics: Applications in Hydrogeology. Cambridge University Press.
3) Armstrong, M. (1998). Basic Linear Geostatistics. Springer.

Type of Course: Major	Economic Geology Course Level: 300 Credit: 3 L -T-P-C: 3-1-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162M602

Course Objectives: The objective of this course is to provide a comprehensive understanding of ore deposits, their formation processes, exploration techniques, resource estimation, and sustainable mineral resource management.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define key concepts of economic geology, such as ores, gangue minerals, tenor, grade, and mineral resources.	**BT 1**
CO 2	Understand the classification of ore deposits based on their mode of formation and describe various ore-forming processes.	**BT 2**
CO 3	Apply geological, geochemical, and geophysical exploration methods for mineral resource assessment.	**BT 3**
CO 4	Analyse mineral deposit models, metallogenic provinces, and global as well as Indian mineral resource distribution.	**BT 4**
CO 5	Evaluate ore grade, reserve estimation methods, and the environmental and economic impacts of resource utilization and mining.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Introduction to Economic Geology** Ores, gangue minerals, tenor, grade, and lodes. Definitions of mineral occurrence, mineral deposit, and ore deposit. Economic and academic classification of mineral resources and reserves. Classification of ore deposits based on mode of formation (concordant and discordant bodies).	12
Unit 2	**Processes of Ore Formation and Deposit Types** Endogenous processes: Magmatic concentration, skarns, greisens, hydrothermal deposits. Exogenous processes: Weathering products and residual deposits, oxidation and supergene enrichment, placer deposits. Metallogenic provinces and epochs (Global and Indian perspective).	12
Unit 3	**Mineral Exploration and Resource Estimation** Exploration techniques: Remote sensing, GIS, geophysical, and geochemical methods. Recent advancements: AI and machine learning in mineral exploration. Ore grade assessment and reserve estimation. Economic and environmental considerations in mineral resource utilization.	12
Unit 4	**Important Economic Minerals of India** Metallic mineral deposits: Iron, copper, manganese, lead, zinc, aluminium, chromium and rare metals and rare earth elements (REE). Non-metallic and industrial minerals: Limestone, phosphate, graphite, bauxite. Atomic minerals: Uranium, thorium bearing minerals. Gemstones: Occurrence and economic significance. Concept of strategic, critical and essential minerals, National Mineral Exploration Trust (NMET), National Critical Mineral Mission (NCMM) Sustainable mining practices and environmental impact assessment.	12
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		42
Total Notional Credit Hours		**90**

Text Books suggested:

1) Evans, A. M. (1993). Ore Geology and Industrial Minerals: An Introduction. Blackwell Science.
2) Banerjee, D. K. (2014). Mineral Resources of India. Narosa Publishing.

Reference Books:

1) Guilbert, J. M., & Park, C. F. (2007). The Geology of Ore Deposits. Waveland Press.
2) Robb, L. J. (2005). Introduction to Ore-Forming Processes. Wiley-Blackwell.
3) Ridley, J. (2013). Ore Deposit Geology. Cambridge University Press.
4) Mookherjee, A. (2000). Ore Genesis: A Holistic Approach. Allied Publishers.

Type of Course: Major	Economic Geology Practical Course Level: 300 Credit: 1 L -T-P-C: 0-1-2-1 Scheme of Evaluation: Practical	Course Code: GEOL162M613

Course Objectives: The objective of this course is to provide a comprehensive understanding of ore deposits, their formation processes, exploration techniques, resource estimation, and sustainable mineral resource management.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define key concepts of economic geology, such as ores, gangue minerals, tenor, grade, and mineral resources.	**BT 1**
CO 2	Understand the classification of ore deposits based on their mode of formation and describe various ore-forming processes.	**BT 2**
CO 3	Apply geological, geochemical, and geophysical exploration methods for mineral resource assessment.	**BT 3**
CO 4	Analyse mineral deposit models, metallogenic provinces, and global as well as Indian mineral resource distribution.	**BT 4**
CO 5	Evaluate ore grade, reserve estimation methods, and the environmental and economic impacts of resource utilization and mining.	**BT 5**

Modules	Topics and Course Content	Hours
I	Study of physical properties (colour, streak, hardness, specific gravity, etc.) of: • **Metallic ores**: Iron (Hematite, Magnetite, Limonite), Copper (Chalcopyrite, Malachite), Manganese (Pyrolusite, Psilomelane), Lead and Zinc (Galena, Sphalerite), Aluminium (Bauxite), Chromium (Chromite). • **Industrial minerals**: Limestone, Fluorite, Graphite, Phosphate, Bauxite, and REE-bearing minerals. Microscopic study of ore-forming minerals (Under Reflected Light): Oxides and Sulphides. Assessment of ore grade and reserve estimation using Geometrical and Geostatistical methods. Preparation of mineral distribution maps of India. Understanding satellite imagery and spectral signatures of ore deposits. **Field visit to underground or open cast mines.**	30
Total Notional Credit Hours		**30**

Text Books suggested:

1) Evans, A. M. (1993). Ore Geology and Industrial Minerals: An Introduction. Blackwell Science.
2) Guilbert, J. M., & Park, C. F. (2007). The Geology of Ore Deposits. Waveland Press.
3) Banerjee, D. K. (2014). Mineral Resources of India. Narosa Publishing.

Reference Books:

1) Robb, L. J. (2005). Introduction to Ore-Forming Processes. Wiley-Blackwell.
2) Craig, J. R., Vaughan, D. J., & Skinner, B. J. (2011). Resources of the Earth: Origin, Use, and Environmental Impact. Prentice Hall.
3) Moon, C. J., Whateley, M. K. G., & Evans, A. M. (2006). Introduction to Mineral Exploration. Blackwell Publishing.
4) Klemm, L., & Heinrich, C. (2019). Economic Geology: Principles and Practice. Wiley-Blackwell.
5) Ridley, J. (2013). Ore Deposit Geology. Cambridge University Press.
6) Mookherjee, A. (2000). Ore Genesis: A Holistic Approach. Allied Publishers.

Type of Course: Major	Engineering and Environmental Geology Course Level: 300 Credit: 3 L -T-P-C: 3-1-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162M604

Course Objectives: To provide students with fundamental knowledge of engineering and environmental geology, focusing on the geological factors influencing construction projects, natural hazards, environmental impact assessment, and sustainable resource management.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall fundamental concepts of engineering geology, rock and soil properties, and environmental geology.	**BT 1**
CO 2	Explain geological considerations in civil engineering projects, such as dams, tunnels, and foundations.	**BT 2**
CO 3	Apply geotechnical classification systems (RQD, RMR, Q-System) for site evaluation in construction projects.	**BT 3**
CO 4	Analyse the causes, impacts, and mitigation strategies for landslides, floods, earthquakes, and water pollution.	**BT 4**
CO 5	Evaluate engineering and environmental geological data for sustainable planning and environmental impact assessment.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Fundamentals of Engineering and Environmental Geology** Scope and significance of Engineering and Environmental Geology. Role of geologists in infrastructure development, urban planning, and environmental management. Engineering properties of soils: Atterberg limits, shear strength, consolidation. Clay minerals and their significance in geotechnical studies.	12
Unit 2	**Geological and Geotechnical Investigations for Engineering Projects** Rock Mass Classification: Rock Quality Designation (RQD), Rock Mass Rating (RMR), Tunnelling Quality Index (Q-System). Geological, geotechnical, and environmental considerations for dams and tunnels. Foundation and abutment Problems and remediation: Grouting, rock bolting, and other stabilization measures.	12
Unit 3	**Environmental Geology and Natural Hazards** Definition, scope, and principles of environmental geology. Pollution due to mining activities, industrial operations, and radioactive mineral extraction. Soil erosion and degradation: Causes and preventive measures. Floods: Causes, impacts, flood frequency analysis, management strategies, and case studies (including Assam). Water pollution: Sources, types, impact on ecosystems, pollution parameters, and potable water standards (Indian and WHO). Landslides: Causes, factors, hazard zonation, and mitigation measures. Earthquakes: Causes, seismic hazard assessment, and engineering solutions.	12
Unit 4	**Environmental Management and Impact Assessment** Introduction to Environmental Impact Assessment (EIA), Environmental Management Systems (EMS) and Environmental Management Plan (EMP). Geological considerations in EIA for large-scale urban and industrial projects. Sustainable resource management and environmental protection strategies.	12
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Notional Credit Hours		**70**

Text Books suggested:

1) Krynine & Judd (2005) – Principles of Engineering Geology and Geotechnics, CBS Publishers.
2) Bell, F.G. (2007) – Fundamentals of Engineering Geology, Butterworth-Heinemann.

Reference Books:

1) Reddy, D.V. (2010) – Engineering Geology for Civil Engineers, Oxford University Press.
2) Gokhale, K.V.G.K. & Rao, D.M. (2018) – Experiments in Engineering Geology, CBS Publishers.
3) Keller, E.A. (2011) – Environmental Geology, Pearson Education.
4) Montgomery, C. (2016) – Environmental Geology, McGraw-Hill.
5) Blyth, F.G.H. & de Freitas, M.H. (1984) – A Geology for Engineers, Butterworth-Heinemann.

Type of Course: Major	Engineering Geology Practical Course Level: 300 Credit: 1 L -T-P-C: 0-1-2-1 Scheme of Evaluation: Practical	Course Code: GEOL162M615

Course Objectives: To provide students with fundamental knowledge of engineering and environmental geology, focusing on the geological factors influencing construction projects, natural hazards, environmental impact assessment, and sustainable resource management.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall fundamental concepts of engineering geology, rock and soil properties, and environmental geology.	**BT 1**
CO 2	Explain geological considerations in civil engineering projects, such as dams, tunnels, and foundations.	**BT 2**
CO 3	Apply geotechnical classification systems (RQD, RMR, Q-System) for site evaluation in construction projects.	**BT 3**
CO 4	Analyse the causes, impacts, and mitigation strategies for landslides, floods, earthquakes, and water pollution.	**BT 4**
CO 5	Evaluate engineering and environmental geological data for sustainable planning and environmental impact assessment.	**BT 5**

Modules	Topics and Course Content	Hours
I	**Measurement of Atterberg Limits (LL, PL, PI) of Soil Samples** – Determination and interpretation of consistency limits and plasticity index. **Determination of Shear Strength of Soils (Demonstration/Data-based)** – Using direct shear test data to evaluate soil strength parameters. **Rock Mass Characterization Exercises** – Calculation and interpretation of Rock Quality Designation (RQD), Rock Mass Rating (RMR), and Tunnelling Quality Index (Q-System) from field/lab datasets. **Preparation and Interpretation of Geological Maps for Engineering Sites** – Identification of suitable and unsuitable zones for dam, tunnel, and bridge construction based on lithology and structure. **Geotechnical Logging of Borehole Data (Core Log Interpretation)** – Assessment of core recovery, RQD, and identification of problematic rock/soil layers. **Identification of Expansive Clay Minerals (e.g., Montmorillonite, Kaolinite)** – Hand specimen or slide/photo identification; interpretation of their engineering implications. **Computation of reservoir area, catchment area, reservoir capacity, and reservoir life.** **Interpretation of geological cross-sections for engineering projects.** **Computation of index properties of rocks.** **Field visit and geological mapping of engineering project sites (dams/tunnels)**	30
Total Notional Credit Hours		**30**

Text Books suggested:

1) Krynine & Judd (2005) – Principles of Engineering Geology and Geotechnics, CBS Publishers.
2) Bell, F.G. (2007) – Fundamentals of Engineering Geology, Butterworth-Heinemann.

Reference Books:

1) Reddy, D.V. (2010) – Engineering Geology for Civil Engineers, Oxford University Press.
2) Gokhale, K.V.G.K. & Rao, D.M. (2018) – Experiments in Engineering Geology, CBS Publishers.
3) Keller, E.A. (2011) – Environmental Geology, Pearson Education.
4) Montgomery, C. (2016) – Environmental Geology, McGraw-Hill.
5) Blyth, F.G.H. & de Freitas, M.H. (1984) – A Geology for Engineers, Butterworth-Heinemann.

Type of Course: Major	Fuel Geology Course Level: 300 Credit: 3 L -T-P-C: 3-1-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162M606

Course Objectives: To provide an in-depth understanding of the geological, geochemical, and economic aspects of conventional and unconventional fuels, including coal, petroleum, natural gas, gas hydrates, and nuclear energy resources, along with their exploration, exploitation, and environmental implications.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define and explain the origin, classification, and fundamental properties of coal, petroleum, and nuclear fuels.	**BT 1**
CO 2	Describe the processes of petroleum generation, migration, and accumulation, as well as the role of kerogen maturation in hydrocarbon formation.	**BT 2**
CO 3	Identify and classify hydrocarbon traps, reservoir rocks, and cap rocks using geological and petrophysical parameters.	**BT 3**
CO 4	Evaluate the feasibility of coal bed methane, gas hydrates, and shale gas as alternative energy resources based on geological and geochemical conditions.	**BT 4**
CO 5	Assess the environmental impact of fuel extraction and suggest sustainable mitigation strategies in the context of modern energy demands.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Coal Geology** Definition, origin, and formation of coal. Coal classification based on rank, type, and grade (Indian and International systems). Fundamentals of Coal Petrology – Lithotypes, microlithotypes, and macerals. Proximate and Ultimate analysis of coal. Coal Bed Methane (CBM): Global and Indian scenario, exploration and production. Underground coal gasification and Coal liquefaction: Principles, techniques, and applications. Environmental impact of coal mining and its mitigation strategies.	12
Unit 2	**Petroleum Geology** Composition and physical properties of crude oil and natural gas. Origin and formation of petroleum – Biogenic and thermal transformation of organic matter. Migration and accumulation of petroleum. Kerogen maturation and classification. Reservoir rocks – Petrophysical properties and classification (clastic vs. chemical). Cap rocks: Definition, properties, and types.	12
Unit 3	**Petroleum Reservoirs and Traps** Hydrocarbon traps: Definition and theories (Anticlinal theory, modern trap theory). Classification of traps – Structural, Stratigraphic, and Combination traps. Time of trap formation and hydrocarbon accumulation. Plate tectonics and global distribution of hydrocarbon reserves. Unconventional hydrocarbons – Oil sands, Oil shale, and Shale gas.	12
Unit 4	**Alternative and Nuclear Fuels** Gas Hydrates: Occurrence, structure, geological settings, stability, and economic significance. Shale Gas and Tight Oil: Exploration, production techniques, and challenges. Nuclear Fuels: • Uranium and Thorium-bearing minerals. • Geochemistry of U and Th. • Metallogenic provinces of India. • Methods of radioactive mineral exploration. • Detectors of radioactivity: Geiger-Müller counter, proportional counter, scintillation counters, and spectrometers.	12
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		24
Total Notional Credit Hours		**90**

Text Books suggested:

1) Thomas, L. (2002). Coal Geology. Wiley-Blackwell.
2) Bjorlykke, K. (2010). Petroleum Geoscience: From Sedimentary Environments to Rock Physics. Springer.
3) IAEA Publications on nuclear fuel exploration and processing.

Reference Books:

1) Gluyas, J., & Swarbrick, R. (2004). Petroleum Geoscience. Blackwell.
2) Tissot, B. P., & Welte, D. H. (1984). Petroleum Formation and Occurrence. Springer.
3) North, F. K. (1985). Petroleum Geology. Springer.
4) Boggs, S. (2006). Principles of Sedimentology and Stratigraphy. Pearson.
5) Merrill, R. K. (1991). Sourcebook for Petroleum Geology. AAPG.
6) Sharma, S. (2017). Environmental Impact of Mining and Mineral Processing. Elsevier.

Type of Course: Major	Fuel Geology Practical	Course Code:
	Course Level: 300 Credit: 1 L -T-P-C: 0-1-2-1 Scheme of Evaluation: Practical	GEOL162M617

Course Objectives: To provide an in-depth understanding of the geological, geochemical, and economic aspects of conventional and unconventional fuels, including coal, petroleum, natural gas, gas hydrates, and nuclear energy resources, along with their exploration, exploitation, and environmental implications.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define and explain the origin, classification, and fundamental properties of coal, petroleum, and nuclear fuels.	**BT 1**
CO 2	Describe the processes of petroleum generation, migration, and accumulation, as well as the role of kerogen maturation in hydrocarbon formation.	**BT 2**
CO 3	Identify and classify hydrocarbon traps, reservoir rocks, and cap rocks using geological and petrophysical parameters.	**BT 3**
CO 4	Evaluate the feasibility of coal bed methane, gas hydrates, and shale gas as alternative energy resources based on geological and geochemical conditions.	**BT 4**
CO 5	Assess the environmental impact of fuel extraction and suggest sustainable mitigation strategies in the context of modern energy demands.	**BT 5**

Modules	Topics and Course Content	Hours
I	Megascopic identification of coal types (Peat, Lignite, Bituminous, Anthracite). Study of coal lithotypes and macerals under a reflected light microscope. Proximate analysis of coal: Moisture content, Volatile matter, Ash content, and Fixed carbon determination, rank and grade determination. Estimation of calorific value of coal using the Dulong formula. Study of physical properties of crude oil (viscosity, density, API gravity). Petrophysical properties of reservoir rocks (porosity, permeability, fluid saturation). Identification and classification of hydrocarbon traps using geological cross-sections. Preparation of isopach and structure contour maps for petroleum exploration. **Industrial visit like coalfields and oilfields of NE India and laboratory visit.**	30
Total Notional Credit Hours		**30**

Text Books suggested:

1) Thomas, L. (2002). Coal Geology. Wiley-Blackwell.
2) Bjorlykke, K. (2010). Petroleum Geoscience: From Sedimentary Environments to Rock Physics. Springer.
3) IAEA Publications on nuclear fuel exploration and processing.

Reference Books:

1) Gluyas, J., & Swarbrick, R. (2004). Petroleum Geoscience. Blackwell.
2) Tissot, B. P., & Welte, D. H. (1984). Petroleum Formation and Occurrence. Springer.
3) North, F. K. (1985). Petroleum Geology. Springer.
4) Boggs, S. (2006). Principles of Sedimentology and Stratigraphy. Pearson.
5) Merrill, R. K. (1991). Sourcebook for Petroleum Geology. AAPG.
6) Sharma, S. (2017). Environmental Impact of Mining and Mineral Processing. Elsevier.

Type of Course: Minor	Environmental Geology Course Level: 200 Credit: 4 L -T-P-C: 3-1-0-4 Scheme of Evaluation: Theory	Course Code: GEOL162N601

Course Objectives: To introduce students to the interactions between geological processes and the environment, focusing on natural hazards, resource management, pollution control, and sustainable development.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall fundamental concepts of environmental geology and natural hazards.	**BT 1**
CO 2	Explain the impact of geological processes on the environment and human activities.	**BT 2**
CO 3	Apply geoscientific methods to assess environmental pollution and resource management.	**BT 3**
CO 4	Evaluate case studies on environmental degradation and mitigation strategies.	**BT 4**
CO 5	Critically assess environmental policies and propose sustainable solutions.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Introduction to Environmental Geology** Definition, scope, and significance of environmental geology. Earth systems and human interactions: lithosphere, hydrosphere, atmosphere, and biosphere. Human impact on geological processes and vice versa.	18
Unit 2	**Environmental Pollution** Water Pollution: Sources (natural and anthropogenic), groundwater contamination, and mitigation. Soil Pollution: Causes (mining, industrial waste), impact, and remediation techniques. Air Pollution and Climate Change: Role of geological processes in atmospheric changes. Radioactive Pollution: Sources (nuclear waste, uranium mining), impacts, and control measures.	18
Unit 3	**Environmental Management and Impact Assessment** Introduction to Environmental Impact Assessment (EIA) and Environmental Management Systems (EMS). Geological considerations in EIA for large-scale urban and industrial projects. Sustainable resource management and environmental protection strategies.	18
Unit 4	**Resource Management and Sustainability** Water Resource Management: Groundwater recharge, rainwater harvesting, and conservation. Soil and Land Use Planning: Soil degradation, erosion control, and sustainable agriculture. Mining and Environmental Impact: Environmental effects of mining, EIA, and mine reclamation. Waste Management: Geology in waste disposal, landfills, and recycling. Sustainable Development Goals (SDGs) and geological perspectives.	18
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		48
Total Notional Credit Hours		**120**

Text Books suggested:

1) Montgomery, C.W. (2020). Environmental Geology. McGraw-Hill.
2) Valdiya, K.S. (2013). Environmental Geology: Indian Context. McGraw-Hill.

Reference Books:

1) Bell, F.G. (2021). Environmental and Engineering Geology. CRC Press.
2) Singh, A.K. & Rajamani, V. (2016). Water Resources and Environmental Geology. Springer.
3) Keller, E.A. (2019). Environmental Geology. Pearson.

Detailed Syllabus Of Semester 7

Type of Course: Major	Advanced Structural Geology and Tectonics Course Level: 400 Credit: 3 L -T-P-C: 3-0-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162M701

Course Objectives: To provide an advanced understanding of deformation processes in the Earth's crust, integrating structural geology principles with plate tectonic mechanisms while developing analytical and field-based skills for geological interpretation.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define and recall fundamental concepts of stress, strain, rock deformation, and tectonic structures.	**BT 1**
CO 2	Understand the mechanics of rock deformation and classify different types of folds, faults, joints, and shear zones.	**BT 2**
CO 3	Apply strain analysis techniques and stereographic projections to interpret structural data.	**BT 3**
CO 4	Analyse deformation patterns to differentiate between brittle and ductile processes and their geological significance.	**BT 4**
CO 5	Evaluate the role of structural geology in tectonics, seismic activity, and resource exploration using field and remote sensing data.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Fundamentals of Structural Geology** Introduction to rock mechanics: Stress and strain in rocks, stress tensor, strain tensor, finite and infinitesimal strain. Mohr stress circle and determination of the direction of shear stress. Principal axes of strain; measurement of strain using Flinn's diagram, Fry's method, and other strain markers. Behaviour of rocks under stress: elastic, plastic, brittle, viscous, and visco-elastic responses and their geological significance. Failure criteria: Coulomb's failure criterion, Griffith's theory of fracture. Planar and linear structures in deformed rocks: Cleavage, lineation, foliation and their kinematic significance.	12
Unit 2	**Folding, Faulting, and Jointing** Classification of folds: Ramsay's (1967) and Fleuty's (1964) classifications. Kinematics of folding: buckle folds, shear folds, and flexural slip folds. Determination of shear sense from fold geometry; superposed folding and interference patterns. Boudinage: Morphology, origin, and relationship to folding. Mechanics of faulting: Anderson's theory of faulting and its limitations. Geometry and kinematics of normal, strike-slip, and thrust faults with natural examples. Concept of fault zone weakening, fault reactivation, and seismotectonics. Geometric analysis of joints: Tectonic, columnar, and release joints.	12
Unit 3	**Shear Zones and Lithospheric Deformation** Shear zones: Geometry, kinematics, and classification. Strain analysis in shear zones: Shear sense indicators. Flow behaviour of sheared rocks: Ductile and brittle-ductile shear zones. Shear zone rocks: Cataclasite, gouge, breccia, mylonite, pseudotachylyte. Microstructures and their significance. Role of shear zones in the evolution of the continental crust.	12

Unit 4	**Tectonics and Structural Applications** Lithospheric plates, plate boundaries, and associated deformation. Orogeny and mountain-building processes: Himalayan tectonics, Andean-type orogeny. Subduction zones, mid-ocean ridges, and transform faults. Concept of Thrust-Tectonics. Triple Junctions and their stability criteria. Concept of spherical motion and Euler's pole in plate tectonics. Tectonic significance of structural geology in earthquake generation, magmatism, and basin evolution. Applications of structural geology in petroleum geology, mineral exploration, and engineering geology. Integration of remote sensing and GIS in structural geology.	12
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		42
Total Credit Hours		**90**

Text Books:

1) Structural Geology - Robert J. Twiss & Eldridge M. Moores, (2nd edition, 2007), W. H. Freeman & Co Ltd.
2) Structural Geology – Haakon Fossen, (2010), Cambridge University Press, New York.
3) Structural Geology- Fundamentals & Modern Developments (1993) - S K Ghosh, Pergamon Press.

Reference Books:

1) Pluijim, B.A.V.D. and Marshak, S., 2003: Earth Structure; 2nd edn., W.W. Norton & Co.
2) Pollard, D.D.,2005: Fundamentals of Structural Geology; Cambridge Univ. Press.

Type of Course: Major	Structural Geology Practical - II Course Level: 400 Credit: 1 L -T-P-C: 0-0-2-1 Scheme of Evaluation: Practical	Course Code: GEOL162M712

Course Objectives: To provide an advanced understanding of deformation processes in the Earth's crust, integrating structural geology principles with plate tectonic mechanisms while developing analytical and field-based skills for geological interpretation.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define and recall fundamental concepts of stress, strain, rock deformation, and tectonic structures.	**BT 1**
CO 2	Understand the mechanics of rock deformation and classify different types of folds, faults, joints, and shear zones.	**BT 2**
CO 3	Apply strain analysis techniques and stereographic projections to interpret structural data.	**BT 3**
CO 4	Analyse deformation patterns to differentiate between brittle and ductile processes and their geological significance.	**BT 4**
CO 5	Evaluate the role of structural geology in tectonics, seismic activity, and resource exploration using field and remote sensing data.	**BT 5**

Modules	Topics and Course Content	Hours
I	**Construction and interpretation of Mohr's stress circle** From given stress tensor data. Identification of principal stress directions and shear stress magnitudes. **Measurement of finite strain using strain markers** Fry's method and Rf/φ method using stretched clasts or reduction spots. **Plotting and interpretation using Flinn Diagram** Strain data plotting and assessment of flattening vs. constrictional strain. **Analysis and classification of folds using Ramsay's and Fleuty's methods** Based on dip/strike data and fold profile geometries. **Kinematic analysis of folds** Determination of shear sense from asymmetric folds and parasitic folds. **Recognition and interpretation of superposed folding** Drawing and interpretation of interference patterns (Type 0, 1, 2, 3). **Stress analysis using fault-slip data** Identification of stress regime (normal, thrust, strike-slip). **Geometric analysis of joint sets** Interpretation of tectonic, cooling, and release joints. **Microscopic and mesoscopic analysis of shear sense indicators** S-C fabrics, σ-clasts, δ-clasts, mica fish (thin section/photo-based exercises). **Euler pole and spherical plate motion exercises** Plotting relative plate motions and angular displacements. **Case study analysis: Himalayan tectonics** Structural cross-sections, lineaments, and tectonic evolution using maps. **Preparation of rose diagrams and stereographic projections** Analysis of joint/fault orientation data using software or manual methods.	30
Total Credit Hours		**30**

Text Books:

1) Structural Geology - Robert J. Twiss & Eldridge M. Moores, (2nd edition, 2007), W. H. Freeman & Co Ltd.
2) Structural Geology – Haakon Fossen, (2010), Cambridge University Press, New York.
3) Structural Geology- Fundamentals & Modern Developments (1993) - S K Ghosh, Pergamon Press.

Reference Books:

1) Pluijim, B.A.V.D. and Marshak, S., 2003: Earth Structure; 2nd edn., W.W. Norton & Co.
2) Pollard, D.D.,2005: Fundamentals of Structural Geology; Cambridge Univ. Press.

Type of Course: Major	Advanced Igneous and Metamorphic Petrology Course Level: 400 Credit: 3 L -T-P-C: 3-0-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162M703

Course Objectives: To provide an in-depth understanding of the genesis, evolution, and geodynamic implications of igneous and metamorphic rocks using petrographic, geochemical, and thermodynamic approaches.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall fundamental igneous and metamorphic processes, including magma generation, crystallisation, and metamorphic transformations.	**BT 1**
CO 2	Understand the geochemical, mineralogical, and textural characteristics of igneous and metamorphic rocks in different tectonic settings.	**BT 2**
CO 3	Application of phase diagrams, geochemical data, and petrological concepts to determine the petrogenesis of igneous and metamorphic rocks.	**BT 3**
CO 4	Analyse various igneous and metamorphic processes by evaluating mineral assemblages, textures, and geochemical trends.	**BT 4**
CO 5	Evaluate the tectonic and thermal evolution of the lithosphere using petrological and geochemical evidence from natural rock samples.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Igneous Processes and Geochemical Characterisation** Partial melting, magma differentiation, and source characterisation. Mantle melting and melt-mantle interaction in different geodynamic settings. Magmatism in mid-ocean ridges, subduction zones, continental and oceanic rift zones, and plume-related settings (hotspots). Major, trace, and isotopic geochemistry in petrogenetic interpretations. Trace element partitioning during equilibrium and fractional crystallisation/melting. Modelling trace element distribution in igneous petrogenesis.	12
Unit 2	**Petrology and Petrogenesis of Igneous Rocks** **Petrology and tectonic significance of major igneous rock types:** Ultramafic rocks (Komatiite, Kimberlite) Ophiolites and layered mafic-ultramafic complexes Alkaline rocks and carbonatites Flood basalts (Deccan Traps, Sylhet Traps) Granitoids and anorthosites – Tectonic discrimination of granitoids, their role in crustal evolution. **Experimental petrology and phase equilibria:** Two-, three-, and four-component phase systems at different pressures and temperatures. **Radiometric dating of igneous rocks and crustal evolution.**	12
Unit 3	**Metamorphic Processes, Reactions, and Textures** Heat flow and Metamorphism Crustal thickening and geothermal gradient. Metamorphic zones, metamorphic grade. Paired metamorphic belts and their plate tectonic significance. Thermodynamic basis of metamorphic facies. Mineral assemblages and phase diagrams (ACF, AKF, AFM). Metasomatism and fluid-rock interactions. **Metamorphic textures and microstructures:** High-strain textures, deformation fabrics, reaction rims, and replacement textures. Analysis of poly-metamorphic and poly-deformed rocks, Metamorphic P-T-t paths	12

Unit 4	**Thermodynamics and Advanced Metamorphic Petrology** **Thermodynamics in metamorphism:** Fundamental thermodynamic equations, enthalpy, entropy, and activity, Gibbs Free Energy. Application of the Clausius-Clapeyron equation in metamorphic reactions. Chemical potential and equilibrium in metamorphic systems. **Solution behaviour in metamorphic minerals:** Mixing components, ideal and non-ideal solutions. Raoult's Law and Henry's Law. **Geothermobarometry and petrogenetic grids:** P-T estimates using exchange and net-transfer reactions. Application of mineral chemistry in deciphering metamorphic conditions. **Role of fluids in metamorphism:** Fluid inclusions and their significance in metamorphic petrology. Retrograde metamorphism and re-equilibration.	12
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		42
Total Credit Hours		**90**

Text Books:

1) Best, M.G. – Igneous and Metamorphic Petrology
2) Philpotts & Ague – Principles of Igneous and Metamorphic Petrology

Reference Books:

1) Winter, J.D. – Principles of Igneous and Metamorphic Petrology
2) Turner, F.J. – Metamorphic Petrology
3) Yardley, B.W.D. – Introduction to Metamorphic Petrology
4) Rollinson, H. – Using Geochemical Data: Evaluation, Presentation, Interpretation

Type of Course: Major	Igneous and Metamorphic Practical Course Level: 400 Credit: 1 L -T-P-C: 0-0-2-1 Scheme of Evaluation: Practical	Course Code: GEOL162M714

Course Objectives: To provide an in-depth understanding of the genesis, evolution, and geodynamic implications of igneous and metamorphic rocks using petrographic, geochemical, and thermodynamic approaches.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall fundamental igneous and metamorphic processes, including magma generation, crystallisation, and metamorphic transformations.	**BT 1**
CO 2	Understand the geochemical, mineralogical, and textural characteristics of igneous and metamorphic rocks in different tectonic settings.	**BT 2**
CO 3	Application of phase diagrams, geochemical data, and petrological concepts to determine the petrogenesis of igneous and metamorphic rocks.	**BT 3**
CO 4	Analyse various igneous and metamorphic processes by evaluating mineral assemblages, textures, and geochemical trends.	**BT 4**
CO 5	Evaluate the tectonic and thermal evolution of the lithosphere using petrological and geochemical evidence from natural rock samples.	**BT 5**

Modules	Topics and Course Content	Hours
I	**Igneous Petrology Practical** 1. Use of phase diagrams (binary and ternary systems) to interpret magma crystallisation trends. 2. Geochemical analysis for petrogenetic interpretation – Use of major and trace element data to determine palaeotectonic settings of igneous rocks. 3. Numerical and graphical problems on magma dynamics: o Solid-liquid equilibrium system. o Magma viscosity and ascent rate. o Fractional crystallisation, partial melting, assimilation, and magma mixing (petrogenetic modelling).	10
II	**Metamorphic Petrology Practical** 4. Recognition of reaction textures, porphyroblastic growth, and deformation features through microscopic study of metamorphic rocks. 5. Identification of index minerals and determination of metamorphic grade. 6. Use of phase diagrams to interpret P-T conditions of metamorphism (ACF, AKF, AFM diagrams). 7. Geothermobarometry calculations using mineral assemblages – Estimation of pressure-temperature conditions of metamorphic reactions. 8. Petrofabric analysis of metamorphic rocks – Measurement of preferred orientation of minerals. 9. Determination of strain and deformation history – Study of deformation textures and structural evolution. 10. Measurement of foliation and lineation orientations – Use of stereographic projections.	10
III	**Fieldwork: Field visit for a period of 1 week from 1st to 7th November** 11. Field study in an igneous and metamorphic terrain – • Identification of primary and secondary igneous structures. • Recognition of metamorphic facies, field textures, and structural fabrics. • Sample collection for petrographic and geochemical analysis.	10
Total Credit Hours		**30**

Text Books:

1) Best, M.G. – Igneous and Metamorphic Petrology
2) Philpotts & Ague – Principles of Igneous and Metamorphic Petrology

Reference Books:

1) Winter, J.D. – Principles of Igneous and Metamorphic Petrology
2) Turner, F.J. – Metamorphic Petrology

3) Yardley, B.W.D. – Introduction to Metamorphic Petrology
4) Rollinson, H. – Using Geochemical Data: Evaluation, Presentation, Interpretation

Type of Course: Major	Advanced Sedimentology and Quaternary Geology Course Level: 400 Credit: 3 L -T-P-C: 3-0-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162M705

Course Objectives: To equip students with an in-depth understanding of sedimentary processes, depositional environments, and Quaternary geological changes, enabling them to reconstruct past climates, sea-level fluctuations, and human-environment interactions through applied geoscientific methods.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define sedimentary textures, structures, and classification schemes.	**BT 1**
CO 2	Explain sedimentary rock classification, textures, and structures and their significance in reconstructing depositional environments.	**BT 2**
CO 3	Identify and differentiate various depositional environments using sedimentological and stratigraphic principles.	**BT 3**
CO 4	Analyse sedimentary basins using sequence stratigraphy and facies models.	**BT 4**
CO 5	Evaluate Quaternary geological changes, including glacial-interglacial cycles, climate proxies, and sea-level fluctuations.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Sedimentary Petrology, Textures, and Geochemistry** **Textures & Structures:** Measurement and statistical interpretation of grain size, shape, and fabric. Genesis and significance of primary sedimentary structures; paleocurrent analysis. **Clastic Sedimentology:** Mineralogy and genetic classification of clastics (sandstone, shale, conglomerates). Fundamentals of clay sedimentology. **Carbonate Sedimentology:** Origin and mineralogy of limestones; Folk and Dunham classification schemes. **Diagenesis:** Processes of lithification in clastics; diagenesis and neomorphism in carbonate rocks. **Geochemistry:** Use of geochemical plots to determine chemical maturity, weathering indices, sediment classification, provenance and tectonic settings.	12
Unit 2	**Depositional Systems, Tectonics, and Stratigraphy** **Facies & Environments:** Continental, transitional, and marine depositional systems. Lithofacies assemblages and Walther's Law of Facies Succession. **Provenance:** Determination of source rocks using heavy minerals, quartz, feldspars, and lithic fragments. **Basins & Tectonics:** Classification of sedimentary basins. Tectonic control on sedimentation and the relationship between plate tectonics and sediment accumulation. **Sequence Stratigraphy:** Key concepts, sequence boundaries, systems tracts, and parasequences.	12
Unit 3	**Quaternary Geology & Climate Change** Introduction to Quaternary Period: Time scale and major climate events. Quaternary climates – Milankovitch cycles and climate forcing, eustatic changes. Proxy indicators of paleoclimatic changes - land, ocean and cryosphere (ice core studies). Sea-level fluctuations: Causes, methods of reconstruction, and impact on sedimentation. Palaeosols, loess deposits, and desertification: Indicators of past climate change. Quaternary Stratigraphy – Oxygen Isotope stratigraphy, biostratigraphy and magnetostratigraphy. Defining the Meghalayan Age: The GSSP, the 4.2 ka Event, Impacts of the Meghalayan Drought, Relevance to Modern Climate Change, The Anthropocene Debate.	12

Unit 4	**Applied Quaternary Geology & Geoarchaeology** Dating methods: Radiocarbon dating, U-series, OSL, Cosmogenic nuclides, Amino acid. Quaternary geomorphology: Responses of geomorphic systems to climate, sea level and tectonics on variable time scales in the Quaternary. Human evolution and environmental changes: Archaeological evidence and climate-human interactions. Geoarchaeology case studies from Indian Sub-continent. Quaternary stratigraphy of India– continental records (fluvial, glacial, aeolian, palaeosols and duricrust); marine records; continental-marine correlation of Quaternary record.	12
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		42
Total Credit Hours		**90**

Text Books:

1) Principles of Sedimentology and Stratigraphy – Sam Boggs Jr.
2) Sedimentology and Stratigraphy – Gary Nichols
3) Sedimentary Rocks – F.J. Pettijohn

Reference Books:

1) Introduction to Sedimentology – S. Sengupta
2) Sedimentary Petrology – Maurice E. Tucker
3) Depositional Sedimentary Environments - Reineck and Singh, (1980), Springer – Verlag.
4) Basin Analysis: Principles and Application to Petroleum Play Assessment – Philip A. Allen & John R. Allen

Type of Course: Major	Sedimentology Practical - II Course Level: 400 Credit: 1 L -T-P-C: 0-0-2-1 Scheme of Evaluation: Practical	Course Code: GEOL162M716

Course Objectives: To equip students with an in-depth understanding of sedimentary processes, depositional environments, and Quaternary geological changes, enabling them to reconstruct past climates, sea-level fluctuations, and human-environment interactions through applied geoscientific methods.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define sedimentary textures, structures, and classification schemes.	**BT 1**
CO 2	Explain sedimentary rock classification, textures, and structures and their significance in reconstructing depositional environments.	**BT 2**
CO 3	Identify and differentiate various depositional environments using sedimentological and stratigraphic principles.	**BT 3**
CO 4	Analyse sedimentary basins using sequence stratigraphy and facies models.	**BT 4**
CO 5	Evaluate Quaternary geological changes, including glacial-interglacial cycles, climate proxies, and sea-level fluctuations.	**BT 5**

Modules	Topics and Course Content	Hours
I	**Granulometric analysis and their interpretation.** **Thin Section Petrography of Sandstones and Limestones** **Preparation of Lithologs and Facies Analysis** • Construction of lithologs from vertical sedimentary sections. • Interpretation of facies variations and depositional sequences. **Plotting palaeocurrent data** • Rose diagram preparation and interpretation using unidirectional and bidirectional flow data. **Classification of sedimentary basins from tectonic maps** • Examples: rift basins, foreland basins, passive margins, strike-slip basins. **Identification of key surfaces and systems tracts from stratigraphic columns** • Sequence boundaries, flooding surfaces, maximum flooding surfaces, and parasequences. **Heavy Mineral Separation and Microscopic Study** • Separation techniques (gravity separation, bromoform method, etc.). • Identification of common heavy minerals and their provenance significance.	30
Total Credit Hours		**30**

Text Books:

1) A Practical Approach to Sedimentology - Roy C. Lindholm
2) Atlas of Sedimentary Rocks under the Microscope - A. E. ADAMS, W. S. MACKENZIE and C. GUILFORD
3) Sedimentary Rocks in the Field: A Colour Guide - Dorrik A.V. Stow
4) Heavy Minerals in Colour: Maria A. Mange , Heinz F. W. Maurer
5) Principles of Sedimentology and Stratigraphy – Sam Boggs Jr.
6) Sedimentology and Stratigraphy – Gary Nichols

Reference Books:

1) Introduction to Sedimentology – S. Sengupta
2) Sedimentary Petrology – Maurice E. Tucker
3) Depositional Sedimentary Environments - Reineck and Singh, (1980), Springer – Verlag.
4) Basin Analysis: Principles and Application to Petroleum Play Assessment – Philip A. Allen & John R. Allen

Course Level: Research	Dissertation Research			Course Code:
	L-T-P-C: 0-0-0-4	Credits: 4	Scheme of Evaluation: Project	GEOL162M728

Course Objective: To enable students to conduct minor-level research by identifying a geoscientific problem, applying appropriate methodologies, and interpreting results systematically, fostering independent thinking and research skills.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Understand the fundamentals of research methodology, scientific writing, and data interpretation in geological studies.	**BT 2**
CO 2	Formulate a research problem, collect and analyse geological data using appropriate field and laboratory techniques.	**BT 3**
CO 3	Interpret geological datasets and present scientific arguments logically and coherently.	**BT 4**
CO 4	Critically assess existing literature and integrate findings to support research conclusions.	**BT 5**
CO 5	Develop a well-structured research report/dissertation and present research findings effectively.	**BT 6**

Sl. No.	Research Outline	Timeline
1	**Introduction to Research & Project Planning** • Fundamentals of research methodology in geosciences • Identifying a research problem and defining objectives • Literature review: sources, citation, and referencing (use of software like Mendeley/Zotero) • Research ethics, plagiarism, and scientific integrity	3 Weeks
2	**Methodology & Data Collection** • Selection of appropriate methods for field/laboratory studies • Techniques in geological mapping, remote sensing, GIS, and geophysical/geochemical analyses • Sampling techniques and data acquisition methods • Use of software for data processing (e.g., QGIS, MATLAB, RockWorks, Surfer, ORIGIN)	4 Weeks
3	**Data Analysis & Interpretation** • Statistical and graphical representation of data • Quantitative and qualitative interpretation of results • Comparing findings with existing literature • Error analysis and uncertainty assessment	4 Weeks
4	**Report Writing & Presentation** • Structure of a research dissertation: Abstract, Introduction, Methodology, Results, Discussion, Conclusion, References • Formatting and writing scientific reports (following university guidelines) • Preparing research posters and PowerPoint presentations • Oral defence and viva-voce examination	4 Weeks
	Total	**15 weeks**

Assessment Criteria:

Component	Marks (%)	Evaluation Criteria
Proposal Submission	10%	Research question, objectives, feasibility
Mid-Term Review	20%	Progress, data collection, preliminary analysis
Final Report	40%	Report quality, analysis, interpretation, presentation
Oral Presentation & Viva	20%	Clarity, content, response to questions
Work Ethics & Engagement	10%	Attendance, effort, interaction with supervisor

Reference Books:

1) Kothari, C. R. (2004). Research Methodology: Methods and Techniques. New Age International.
2) Press, F., & Siever, R. (2001). Understanding Earth. W. H. Freeman.
3) Tucker, M. E. (2011). Sedimentary Rocks in the Field: A Practical Guide. Wiley-Blackwell.
4) Compton, R. R. (1985). Geology in the Field. Wiley.
5) Petrie, E. S., & Boggs, S. (2007). Geological Field Techniques. Wiley.
6) Academic papers & journal articles related to the research topic.

Type of Course: Major	Climatology and Oceanography Course Level: 400 Credit: 4 L -T-P-C: 3-1-0-4 Scheme of Evaluation: Theory	Course Code: GEOL162M707

Course Objectives: To provide an understanding of the Earth's climate system, atmospheric dynamics, oceanic circulation, and their interactions, with a focus on their geological significance and relevance to climate change.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define key atmospheric and oceanographic concepts, terminologies, and classification systems.	**BT 1**
CO 2	Understand the interactions between the atmosphere, hydrosphere, and lithosphere in climate regulation.	**BT 2**
CO 3	Apply climatic and oceanographic data to interpret weather patterns, ocean circulation, and climate variability.	**BT 3**
CO 4	Analyse the causes and consequences of climate change using historical and modern datasets.	**BT 4**
CO 5	Evaluate the impact of human activities on climate and oceanic systems and propose sustainable solutions.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Fundamentals of Climatology** Structure and chemical composition of the atmosphere, lapse rate and stability Solar radiation and Earth's energy budget Atmospheric temperature, pressure, and humidity distribution Cloud formation and precipitation processes Winds and general circulation patterns Jet streams and monsoonal systems w.r.t. to Indian Sub-continent Western disturbances and severe local convective systems Climatic zones and classification (Köppen & Thornthwaite)	18
Unit 2	**Atmospheric Dynamics and Climate Change** Atmospheric turbulence and boundary layer. Atmospheric stability and weather disturbances (cyclones, anticyclones, tornadoes) El Niño, La Niña, and Southern Oscillation (ENSO), Indian Ocean Dipole. Greenhouse effect and global warming Climatic and sea level changes on different time scales. Ice ages and Milankovitch cycles Climate modelling and prediction	18
Unit 3	**Oceanography and Ocean Circulation** Origin and evolution of oceans Physical and chemical properties of seawater (temperature, salinity, density) Residence times of elements in sea water. Oceanic circulation: Surface currents, thermohaline circulation, Coriolis effect and Ekman spiral, convergence, divergence and upwelling. Ocean waves and tides: Formation, classification, and effects Ocean-atmosphere interaction and its role in climate regulation Marine sediments and their significance in paleoceanography	18
Unit 4	**Marine Resources and Oceanic Processes** Oceanic productivity and biological zonation Coral reefs and their geological significance Hydrothermal vents and deep-sea ecosystems Marine pollution and its impact on climate Ocean exploration techniques (Remote Sensing, SONAR, Argo floats) Impact of climate change on oceans (sea-level rise, acidification) Opening and closing of ocean gateways and their effect on Cenozoic circulation and climate.	18
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		48
Total Credit Hours		**120**

Text Books:

1) Barry & Chorley – "Atmosphere, Weather and Climate"
2) Pinet – "Invitation to Oceanography"

Reference Books:

1) Critchfield – "General Climatology"
2) Lutgens & Tarbuck – "The Atmosphere: An Introduction to Meteorology"
3) Garrison – "Oceanography: An Invitation to Marine Science"
4) Trenberth – "Climate System Modeling"
5) Glenn & Turekian – "Oceans"
6) IPCC Reports – "Climate Change Assessments"

Type of Course: Minor	Stratigraphy Course Level: 300 Credit: 4 L -T-P-C: 3-1-0-4 Scheme of Evaluation: Theory	Course Code: GEOL162N701

Course Objectives: To introduce fundamental principles of stratigraphy, correlation techniques, depositional environments, and the geological history of Earth, with an emphasis on Indian stratigraphy.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall fundamental principles of stratigraphy and stratigraphic units.	**BT 1**
CO 2	Explain stratigraphic correlation techniques and depositional environments.	**BT 2**
CO 3	Apply knowledge of stratigraphy to interpret geological history.	**BT 3**
CO 4	Evaluate the impact of sea-level changes, plate tectonics, and extinction events on stratigraphy.	**BT 4**
CO 5	Assess the economic significance of stratigraphic units for resource exploration.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Fundamentals of Stratigraphy** Definition, scope, and significance of stratigraphy. Fundamental principles of stratigraphy. Lithostratigraphy, Biostratigraphy, Chronostratigraphy, and Sequence Stratigraphy. Concept of facies and facies models. Unconformities and their significance in geological history.	18
Unit 2	**Stratigraphic Classification and Correlation** Lithostratigraphic units: Formation, Member, and Group. Biostratigraphy and biozones: Index fossils and their significance. Chronostratigraphy and Geochronology: Eons, Eras, Periods, and Epochs. Methods of stratigraphic correlation: Lithological, palaeontological, and geochronological approaches. Sea-level changes and their impact on stratigraphic records.	18
Unit 3	**Global Stratigraphy and Major Geological Events** Precambrian and Phanerozoic stratigraphy: Evolutionary trends and fossil records. Plate tectonics and its impact on stratigraphic sequences. Major extinction events in Earth's history. Sequence stratigraphy: Systems tracts, Sequence boundaries, and Parasequences. International Stratigraphic Code and Nomenclature.	18
Unit 4	**Indian Stratigraphy and Economic Significance** Precambrian Stratigraphy of India: Dharwar, Aravalli, Singhbhum, and Bundelkhand cratons. Palaeozoic Stratigraphy of India: Fossiliferous sequences of the Himalayas. Mesozoic Stratigraphy of India: Gondwana formations and their economic importance. Cenozoic Stratigraphy of India: Siwaliks, Deccan Traps, and Quaternary deposits. Stratigraphy of North-East India.	18
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		48
Total Credit Hours		**120**

Text Books suggested:

1) Boggs, S. (2020). Principles of Sedimentology and Stratigraphy. Pearson.
2) The Making of India – K. S. Valdiya, Macmillan India Pvt. Ltd. (2010)

Reference Books:

1) Brenchley, P.J. & Harper, D.A.T. (1998). Palaeoecology: Ecosystems, Environments and Evolution. Springer.
2) Krishnan, M.S. (2009). Geology of India and Burma. CBS Publishers.
3) Ramakrishnan, M. & Vaidyanadhan, R. (2008). Geology of India (Vol. I & II). Geological Society of India.

Detailed Syllabus Of Semester 8

Type of Course: Major	Geomorphology Course Level: 400 Credit: 3 L -T-P-C: 3-0-0-3 Scheme of Evaluation: Theory	Course Code: GEOL162M801

Course Objectives: To develop an advanced understanding of geomorphic processes, with a focus on fluvial systems, tectonic influences, and GIS-based spatial analysis for landform evolution and environmental assessment.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall and describe fundamental geomorphic processes, landform development, and associated geological factors.	**BT 1**
CO 2	Describe the fundamental geomorphic processes and their role in landform development.	**BT 2**
CO 3	Apply morphometric techniques to analyse drainage basins and river systems.	**BT 3**
CO 4	Analyse and interpret spatial data using GIS and remote sensing for geomorphological studies.	**BT 4**
CO 5	Evaluate the impact of climate and tectonics on landscape evolution using geomorphic indices.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Advanced Concepts in Geomorphology** Tectonic and climatic controls on landscape evolution. Rates of uplift and denudation; interaction between endogenic and exogenic processes. Models of long-term landscape development. Quaternary climate change: glacial/interglacial cycles, Milankovitch hypothesis, climate records in sediments. Sea-level changes and landscape evolution.	12
Unit 2	**Fluvial and Coastal Geomorphology** Channel geometry and drainage patterns; structural control on fluvial systems. River hydrodynamics: processes of erosion, transportation, and deposition. Drainage basin evolution and morphometry; role of lithology and tectonics. GIS and remote sensing applications in fluvial geomorphology: DEM-based watershed analysis, channel migration studies, and floodplain mapping. Coastal geomorphology: shore zone processes, erosional and depositional landforms. Coastal vulnerability assessment using GIS.	12
Unit 3	**Tectonic Geomorphology** Geomorphic markers of active tectonics (e.g., fault scarps, river anomalies). Geomorphic indices of active tectonics (e.g., stream gradient index, hypsometric integral). River response to climate change and tectonics; river terraces and knickpoints. Relationship between tectonics and drainage evolution.	12
Unit 4	**Applied Geomorphology and Modern Techniques** Mass wasting: classification, triggering mechanisms, and hazard assessment. Application of GIS in landform mapping and change detection. Role of geomorphology in natural hazard assessment (landslides, floods). Remote sensing and terrain analysis for landform studies. Application of geomorphic principles in environmental and engineering projects.	12
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		42
Total Notional Credit Hours		**90**

Text Books:

1) Bloom, A.L. – Geomorphology: A Systematic Analysis of Late Cenozoic Landforms (Prentice Hall)
2) Huggett, R.J. – Fundamentals of Geomorphology (Routledge)
3) Obrien, P. & Pike, R. – Geomorphometry: Concepts, Software, Applications (Elsevier)

Reference Books:

1) Summerfield, M.A. – Global Geomorphology (Longman)
2) Thornbury, W.D. – Principles of Geomorphology (Wiley)
3) Kale, V.S., & Gupta, A. – Introduction to Geomorphology (Universities Press)
4) Schumm, S.A. – River Variability and Complexity (Cambridge University Press)
5) Burbank, D.W. & Anderson, R.S. – Tectonic Geomorphology (Blackwell)
6) Bishop, M.P. & Shroder, J.F. – Remote Sensing and GIS for Natural Hazards Assessment (Taylor & Francis)
7) Montgomery, D.R. & Dietrich, W.E. – Topographic Controls on Watershed-Scale Erosion and Deposition

Type of Course: Major	Geomorphology Practical Course Level: 400 Credit: 1 L -T-P-C: 0-0-2-1 Scheme of Evaluation: Practical	Course Code: GEOL162M812

Course Objectives: To develop an advanced understanding of geomorphic processes, with a focus on fluvial systems, tectonic influences, and GIS-based spatial analysis for landform evolution and environmental assessment.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall and describe fundamental geomorphic processes, landform development, and associated geological factors.	**BT 1**
CO 2	Describe the fundamental geomorphic processes and their role in landform development.	**BT 2**
CO 3	Apply morphometric techniques to analyse drainage basins and river systems.	**BT 3**
CO 4	Analyse and interpret spatial data using GIS and remote sensing for geomorphological studies.	**BT 4**
CO 5	Evaluate the impact of climate and tectonics on landscape evolution using geomorphic indices.	**BT 5**

Modules	Topics and Course Content	Hours
I	**Topographic and Remote Sensing Analysis** Interpretation of landforms using topographical maps, satellite images, and DEMs. Extraction and analysis of watersheds and drainage networks using GIS and remote sensing. Digital Elevation Model (DEM) processing for terrain analysis and slope mapping.	7
II	**Fluvial Geomorphology Exercises** Preparation of longitudinal river profiles using GIS. Calculation of stream length-gradient index, hypsometric integral, and bifurcation ratio. Automated drainage basin delineation using GIS. Mapping and analysing river meandering, channel migration, and floodplain changes using time-series satellite images. Sediment yield estimation and flood hazard zonation using GIS-based models.	10
III	**Tectonic Geomorphology** Application of geomorphic indices (e.g., mountain-front sinuosity, valley floor width-to-height ratio) using GIS. Mapping and quantifying active faulting and neotectonic deformation using remote sensing techniques.	6
IV	**Coastal Geomorphology** Coastal landform mapping and shoreline change analysis using multi-temporal satellite data. **Applied Geomorphology** Landslide susceptibility mapping using GIS-based models (e.g., AHP, Frequency Ratio). Soil erosion modelling using GIS-based RUSLE (Revised Universal Soil Loss Equation). Geomorphological hazard zonation integrating remote sensing and field-based data.	7
Total Notional Credit Hours		**30**

Text Books:

1) Bloom, A.L. – Geomorphology: A Systematic Analysis of Late Cenozoic Landforms (Prentice Hall)
2) Huggett, R.J. – Fundamentals of Geomorphology (Routledge)
3) Obrien, P. & Pike, R. – Geomorphometry: Concepts, Software, Applications (Elsevier)

Reference Books:

1) Summerfield, M.A. – Global Geomorphology (Longman)
2) Thornbury, W.D. – Principles of Geomorphology (Wiley)
3) Kale, V.S., & Gupta, A. – Introduction to Geomorphology (Universities Press)
4) Schumm, S.A. – River Variability and Complexity (Cambridge University Press)
5) Burbank, D.W. & Anderson, R.S. – Tectonic Geomorphology (Blackwell)
6) Bishop, M.P. & Shroder, J.F. – Remote Sensing and GIS for Natural Hazards Assessment (Taylor & Francis)
7) Montgomery, D.R. & Dietrich, W.E. – Topographic Controls on Watershed-Scale Erosion and Deposition

Type of Course: Minor	Research Methodology Course Level: 400 Credit: 4 L -T-P-C: 3-1-0-4 Scheme of Evaluation: Theory	Course Code: GEOL162M803

Course Objectives: To equip students with the fundamental principles of research, methodologies for data collection and analysis, scientific writing skills, and ethical considerations in research.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define key research concepts and methodologies.	**BT 1**
CO 2	Explain different research designs and data collection techniques.	**BT 2**
CO 3	Apply statistical and analytical tools to interpret research data.	**BT 3**
CO 4	Evaluate research findings for accuracy, reliability, and significance.	**BT 4**
CO 5	Develop well-structured research reports, proposals, and presentations.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Introduction to Research** Definition, characteristics, and objectives of research. Types of research: Basic vs. Applied, Qualitative vs. Quantitative. Research process: Identifying a research problem, formulating hypotheses, and setting objectives. Literature review: Sources, referencing styles, and plagiarism detection tools. Research ethics and integrity: Ethical considerations, avoiding bias, and responsible authorship.	18
Unit 2	**Research Design and Methodology** Research design: Experimental, Descriptive, and Analytical approaches. Sampling techniques: Probability and Non-Probability sampling methods. Data collection methods: Field surveys, laboratory experiments, remote sensing, and archival research. Questionnaire design and interviews: Structure, reliability, and validity. Introduction to case study methodology and mixed-methods research.	18
Unit 3	**Data Analysis and Interpretation** Basics of statistical analysis: Measures of central tendency, dispersion, and correlation. Hypothesis testing: Parametric and non-parametric tests, p-values, and confidence intervals. Introduction to software tools: Excel, SPSS, R, and GIS for research. Data visualization: Graphs, charts, and mapping techniques. Errors in research: Accuracy, precision, and reproducibility of results.	18
Unit 4	**Scientific Writing and Presentation** Structure of a research paper: Abstract, Introduction, Methodology, Results, Discussion, and Conclusion. Writing a thesis or dissertation: Formatting, citation styles (APA, MLA, Chicago), and bibliography management. Writing research proposals and funding applications. Presentation skills: Creating effective slides and posters. Publication process: Peer review, impact factor, indexing, and open-access journals.	18
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		48
Total Notional Credit Hours		**120**

Text Books suggested:

1) Kothari, C.R. & Garg, G. (2019). Research Methodology: Methods and Techniques. New Age International.
2) Creswell, J.W. & Creswell, J.D. (2018). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches. Sage Publications.

Reference Books:

1) Krishnaswamy, K.N., Sivakumar, A.I. & Mathirajan, M. (2009). Management Research Methodology: Integration of Principles, Methods, and Techniques. Pearson.
2) Walliman, N. (2017). Research Methods: The Basics. Routledge.
3) Zikmund, W.G., Babin, B.J., Carr, J.C., & Griffin, M. (2016). Business Research Methods. Cengage Learning.

Type of Course: Major	Dissertation Credit: 12 Scheme of Evaluation: Project	Course Code: GEOL162M824

Course Objective: To develop advanced research skills in geological sciences by conducting independent research, applying analytical tools, and effectively communicating scientific findings.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Prepare a scientific research problem and design a feasible methodology.	**BT 2**
CO 2	Conduct advanced field investigations, data collection, and laboratory analyses.	**BT 3**
CO 3	Apply geospatial, statistical, and computational methods to interpret geological datasets.	**BT 4**
CO 4	Critically evaluate results and draw meaningful geological conclusions.	**BT 5**
CO 5	Write a scientific dissertation, including literature review, methodology, results, and discussion.	**BT 6**

Sl. No.	Research Outline	Timeline
1	**Advanced Research Planning & Proposal Writing** ● Selection of research topic and problem formulation. ● Review of scientific literature, gap analysis, and research hypothesis formation. ● Research proposal writing: Objectives, methodology, data requirements. ● Ethical considerations in research (plagiarism, data integrity, authorship).	2 Weeks
2	**Data Collection, Processing & Methodology** ● Field investigations: Geological mapping, sampling, geophysical/geochemical surveys. ● Data collection techniques: Borehole logging, GIS, remote sensing, petrography, geostatistics. ● Experimental methods: XRD, XRF, SEM-EDS, thin section petrography, sediment analysis. ● Computational techniques: Python/R for geosciences, RockWorks, ArcGIS/QGIS applications.	4 Weeks
3	**Analysis, Interpretation & Discussion** ● Data processing & interpretation: Statistical and spatial analysis, cross-validation. ● Conceptual geological models: Structural, hydrogeological, or mineral deposit models. ● Comparison with previous studies & existing theories. ● Scientific discussions: Uncertainty assessment, limitations of findings.	5 Weeks
4	**Report Writing, Publication & Presentation** ● Scientific report structure: Abstract, introduction, methodology, results, discussion, conclusion. ● Formatting as per journal/conference standards. ● Graphical representation: Maps, cross-sections, geospatial models. ● Preparation for oral defence & viva-voce.	4 Weeks
Total		**15 weeks**

Assessment Criteria:

Component	Marks (%)	Evaluation Criteria
Proposal Presentation	10%	Clarity, feasibility, scientific value
Mid-Term Review & Progress Report	20%	Quality of research progress
Dissertation Report	40%	Depth, originality, scientific rigour
Oral Defence & Viva	20%	Presentation skills, depth of understanding

Component	Marks (%)	Evaluation Criteria
Research Ethics & Engagement	10%	Effort, interactions, adherence to research norms

Reference Books:

1) Kothari, C. R. (2004). Research Methodology: Methods and Techniques.
2) Davis, J. C. (2002). Statistics and Data Analysis in Geology.
3) Bonham-Carter, G. (1994). Geographic Information Systems for Geoscientists: Modelling with GIS.
4) Telford, W. M., Geldart, L. P., & Sheriff, R. E. (1990). Applied Geophysics.
5) Montello, D. R., & Sutton, P. C. (2012). An Introduction to Scientific Research Methods in Geography and Environmental Studies.
6) Academic papers & journal articles related to the research topic.

Type of Course: Major	Geology of North-East India Course Level: 400 Credit: 4 L -T-P-C: 3-1-0-4 Scheme of Evaluation: Theory	Course Code: GEOL162M804

Course Objectives: To provide a detailed understanding of the geological framework, tectonics, stratigraphy, structural geology, mineral resources, and natural hazards of Northeast India, integrating recent research and advancements.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall the major lithostratigraphic units, structural features, and mineral resources of Northeast India.	**BT 1**
CO 2	Understand the tectonic evolution and stratigraphic framework of the region in relation to plate tectonics and basin development.	**BT 2**
CO 3	Interpret geological maps, seismic data, and remote sensing information to assess geological hazards and resource potential.	**BT 3**
CO 4	Analyse the impact of tectonics, climate, and human activities on the geological processes of Northeast India.	**BT 4**
CO 5	Assess the economic potential of mineral and hydrocarbon resources and their sustainable utilization.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Tectonic and Structural Framework** Geographical and geological setting of Northeast India Tectonic domains: Shillong Plateau, Indo-Myanmar Orogenic Belt, Eastern Himalayan Syntaxis, Bengal Basin Evolution of the Indian Plate and its interaction with the Eurasian and Burmese plates Seismicity and active tectonics: fault systems, earthquake zones, and paleoseismic studies Geodynamic evolution of the Himalayas, Indo-Burma Ranges, and Assam-Arakan Basin	18
Unit 2	**Stratigraphy and Palaeontology** Precambrian formations: Shillong Group, Gneissic Complex, and associated lithounits Gondwana formations of Northeast India: stratigraphy, sedimentation, and palaeoclimate Mesozoic sequences: Cretaceous-Tertiary boundary and associated intrusions (Sylhet Traps, Abor volcanics, Lichi volcanics), Ultramafic complexes of NE India (Sung, Samchampi, etc.). Cenozoic stratigraphy of the Assam-Arakan Basin. Palaeontology of Northeast India: Characteristic Flora, Fauna and Microfossils (including spores and pollens).	18
Unit 3	**Economic Geology and Hydrocarbon Resources** Mineral resources: coal, limestone, petroleum, uranium, graphite, vanadium, and rare earth elements Oil and gas fields of NE India Coal deposits of NE India Uranium occurrences in Meghalaya and associated radioactive mineralization Hot springs and their geothermal Potential in Northeast India	18
Unit 4	**Geomorphology, Environmental Geology, and Hazards** Geomorphology of the Brahmaputra Valley Drainage characteristics of Brahmaputra, Barak, Subansiri, Lohit River systems Geohazards: seismic hazards, Floods and landslides in Northeast India Case study on major earthquake events of 1897 and 1950 in Northeast India Soil erosion in Northeast India Environmental impact of mining and hydroelectric projects in the region	18
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		48
Total Notional Credit Hours		**120**

Text Books:

1) Krishnan, M.S. (2017) – Geology of India and Burma, CBS Publishers.
2) Valdiya, K.S. (2016) – Himalayan Geology, Springer.
3) Banerjee, A. (2015) – Tectonics of the Eastern Himalayas and Indo-Myanmar Orogenic Belt, Cambridge Scholars Publishing.

Reference Materials:

1) Geological Survey of India (GSI) Memoirs and Publications on Northeast India, Various editions. (Reports & Books)
2) Jain, A.K. & Manickavasagam, R.M. (2018) – Geology of the Himalayas and Northeast India, Elsevier. (Book)
3) Acharyya, S.K. (2007) – Tectonic Framework and Evolution of the Eastern Himalayas and Indo-Burma Orogen, Journal of Asian Earth Sciences, 29(2), 219–233. (Research Paper)
4) Murthy, K.S.R. et al. (2012) – Petroleum Geology of the Assam-Arakan Basin, Journal of Petroleum Geology, 35(4), 321–340. (Research Paper)
5) Rai, S. et al. (2020) – Seismotectonics of Northeast India: Recent Advances, Tectonophysics, 796, 228–245. (Research Paper)

Type of Course: Major	Planetary Geology Course Level: 400 Credit: 4 L -T-P-C: 3-1-0-4 Scheme of Evaluation: Theory	Course Code: GEOL162M805

Course Objective: To develop an understanding of the geological processes shaping planetary bodies, the evolution of the Solar System, and the application of remote sensing and astrobiology in planetary exploration.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Recall key concepts related to planetary formation, surface processes, and geological evolution of Solar System bodies.	**BT 1**
CO 2	Explain the geological features and atmospheric evolution of terrestrial planets, moons, and asteroids.	**BT 2**
CO 3	Apply remote sensing and crater dating methods to interpret planetary surface processes.	**BT 3**
CO 4	Compare planetary environments to assess habitability and geological activity.	**BT 4**
CO 5	Evaluate the feasibility of space resource utilisation and human exploration strategies.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Origin and Evolution of the Solar System** The formation and differentiation of planetary bodies. Origin of elements, planetary accretion, and core formation. Methods of Solar System exploration: space missions, remote sensing, sample return missions. Meteorites, asteroids, and comets as records of early Solar System processes. Giant impacts and planetary evolution (e.g., Earth-Moon system formation). Dating planetary surfaces using crater analysis and radiometric methods.	18
Unit 2	**Comparative Planetary Geology** Thermal evolution of planets and moons: influence of planetary size and composition. Planetary atmospheres: evolution, retention, and climate history. Surface and internal geology of terrestrial planets: **Mercury:** Tectonics, volcanism, and magnetic field. **Venus:** Surface weathering, volcanism, and atmospheric dynamics. **Moon:** Regolith formation, impact cratering, and volcanic plains. **Mars:** Volcanism (Tharsis region), fluvial and glacial geomorphology, dust storms. Geological and geophysical properties of major moons (e.g., Europa, Titan, Ganymede). Giant planets and their satellites: structure, atmospheres, and magnetospheres.	18
Unit 3	**Planetary Surface Processes & Remote Sensing Applications** Impact cratering: formation stages, ejecta distribution, and shock metamorphism. Volcanism on terrestrial planets and icy moons (e.g., cryovolcanism on Enceladus). Aeolian, fluvial, and glacial processes on Mars and Titan. Remote sensing techniques in planetary geology: multispectral imaging, radar, LIDAR. Data analysis from space missions (Lunar Reconnaissance Orbiter, Mangalyaan, Perseverance) Applications of GIS and machine learning in planetary surface mapping.	18
Unit 4	**Astrobiology and Planetary Exploration** Habitability criteria in the Solar System: liquid water, energy sources, and organic molecules. Biosignatures and life-detection strategies on Mars, Europa, and Enceladus. Exoplanetary geology: Earth-like planets and their geological potential. Terraforming Mars: scientific feasibility and challenges. Future planetary exploration missions and their scientific objectives (e.g., Artemis, Dragonfly, Mars Sample Return). Space resources: mining asteroids and lunar regolith for sustainable space exploration.	18
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		48
Total Notional Credit Hours		**120**

Text Books:

1) The New Solar System – Beatty, Petersen, Chaikin (5th Ed., 1999)
2) Planetary Geology – Greeley & Batson (1990)
3) Comparative Planetology, Geological Perspectives – Ronald Greeley (1985)

Reference Materials:

1) Meteorites and Their Parent Planets – Harry Y. McSween (2nd Ed., 1999, Book)
2) Solar System Evolution: A New Perspective – Stuart Ross Taylor (2nd Ed., 2001, Book)
3) Astrobiology: A Very Short Introduction – David Catling (2013, Book)
4) Evidence for Water on Mars – Malin & Edgett (Science, 2000, Paper)
5) Impact Cratering as a Geological Process – Melosh (1989, Paper)
6) Planetary Science: The Science of Planets Around Stars – de Pater & Lissauer (2015, Book)
7) NASA Technical Reports on Planetary Geology and Remote Sensing (Various Years, Reports)

Type of Course: Major	Urban Geology Course Level: 400 Credit: 4 L -T-P-C: 3-1-0-4 Scheme of Evaluation: Theory	Course Code: GEOL162M806

Course Objective: To develop an understanding of geological processes in urban environments and equip students with the knowledge and skills to assess geological hazards, manage urban resources, and apply geotechnical and GIS-based solutions for sustainable urban development.

Course Outcomes	Description	Bloom's Taxonomy
CO 1	Define and explain fundamental concepts of urban geology, including geological factors influencing urbanisation and sustainability.	**BT 1**
CO 2	Describe and analyse various geohazards such as earthquakes, landslides, and subsidence, and assess their impact on urban areas.	**BT 2**
CO 3	Apply geological principles in geotechnical investigations, land-use planning, and hazard mitigation strategies.	**BT 3**
CO 4	Analyse groundwater resources, urban mineral resources, and environmental concerns to develop sustainable management strategies.	**BT 4**
CO 5	Evaluate geological risks using GIS and remote sensing techniques for hazard assessment and urban planning.	**BT 5**

Modules	Topics and Course Content	Hours
Unit 1	**Fundamentals of Urban Geology** Definition, scope, and significance of Urban Geology Geological factors influencing urban development and sustainability Subsurface geological characterisation in urban areas Soil mechanics and foundation engineering for construction Geological materials in urban infrastructure development	18
Unit 2	**Geohazards and Risk Assessment in Urban Areas** Geological hazards: Earthquakes, landslides, floods, subsidence, sinkholes Urban flood risk assessment and mitigation strategies Engineering solutions for urban landslides and slope stability Seismic microzonation and earthquake-resistant design principles Role of remote sensing and GIS in urban hazard mapping	18
Unit 3	**Urban Resources and Environmental Management** Groundwater resources, aquifer characterisation, and management Urban mineral resources: Quarrying, excavation, and impacts Geothermal energy in cities: Potential, challenges, and applications Urban pollution: Sources, groundwater contamination, and remediation Sustainable urban planning: Environmental impact assessments (EIA)	18
Unit 4	**Applied Urban Geology and GIS Applications** Geology-based urban planning and land-use zoning Geological aesthetics and urban greenspace development Use of GIS and remote sensing in urban geology studies Case studies of geological hazards in urban environments Policies and regulations for urban geological sustainability	18
Experiential Learning: Home Assignments – 8 hrs, Presentation – 8 hrs, Video Screening – 8 hrs		48
Total Notional Credit Hours		**120**

Text Books:

1) Bell, F. G. (2004). Engineering Geology and Construction. CRC Press.
2) McCall, G. J. H., Marker, B. R. & Laming, D. J. C. (2004). Urban Geology in Land Use Planning. Geological Society of London.

Reference Materials:

1) Keller, E. A. & DeVecchio, D. E. (2019). Natural Hazards: Earth's Processes as Hazards, Disasters, and Catastrophes. Routledge.
2) Goudie, A. (2018). Human Impact on the Natural Environment: Past, Present, and Future. Wiley-Blackwell.
3) Yan, J. & Edwards, P. (2019). GIS and Geostatistical Techniques for Groundwater Science. Elsevier.
4) Rivas, V. & Horacio, D. (2014). Urban Geology in Latin America. Springer.
5) van Westen, C. J. (2000). GIS in Natural Hazard Assessment. ITC Journal, 2(3), 45-58. [Paper]
6) Brunsden, D. (1993). The Role of Geomorphology in Urban Planning and Hazard Assessment. Geological Society Special Publications, 14, 63-75. [Paper]